

Article

Artificial Intelligence in Emergency Medical Services for Disasters and Health Emergencies: A Systematic Review

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Abstract

Background: In response to the ever-increasing complexity of health emergencies, the use of intelligent decision-support tools has become essential for effective decision-making within healthcare systems. Among these tools, Artificial Intelligence (AI) stands out for its ability not only to mimic human reasoning but also to process large amounts of information with significant variability. Through formal methods, AI offers a promising solution to address the meta-uncertainty inherent in both health emergency situations and routine healthcare operations. Within this context, pre-hospital care systems play a pivotal role in mitigating the risks associated with health emergencies, as they function as the first point of contact between the population and the healthcare system. **Methods:** A systematic review was conducted across PubMed, IEEE Xplore, Web of Science, and Embase following PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines. A time filter was applied, limiting the selection of articles to publications between 2015 and 2025. The inclusion and exclusion criteria were defined, verified, and then validated by peers. To be included, a study had to be open access and fall within the following fields: AI, Emergency Medical Service (EMS), forecasting, health emergencies, public health, or disaster management. Cohort, comparative, or retrospective studies, as well as those not using real-world data, were excluded. **Results:** The application of the PRISMA methodology on queried articles resulted in 138. Analysis of a cloud of key information extracted from titles, keywords, and abstracts, developed using natural language processing techniques, LiteRev and clustering methods reveals the need to structure the literature review around thematic such as: AI applications for predicting epidemics and natural disasters — in-depth exploration of the uses of AI in emergency medical services — the AI algorithms used — their complexity — explainability, interpretability, and uncertainty quantification. **Conclusion:** This study revealed the pivotal role of AI in forecasting and preparing responses to health emergencies. It represents an essential tool for transforming pre-hospital care systems, which play a key role in responding to such events. However, although AI is widely adopted and has demonstrated spectacular progress, it still faces several challenges, including data incompleteness, representativeness, model interpretability and explainability, inability to quantify uncertainties, as well as human and ethical safeguards.

Received:

Revised:

Accepted:

Published:

Citation: Kokou, L. E., Larribau, R., Orel, E., Elfahim, O., Cossus, J., Yendouname, K., Vaussenat, F., Douglas T., Robert-Nicoud, S., Barakat, O., & Desmettre, T. Artificial Intelligence in Emergency Medical Services for Disasters and Health Emergencies: A Systematic Review. *Journal Not Specified* **2025**, *1*, 0. <https://doi.org/>

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Keywords: Artificial Intelligence; Uncertainty; Health Emergencies; Disasters; Forecasting.

1. Introduction

Over the past decades, health emergencies and natural disasters have multiplied world-wide, with increasing frequency and severity [1]. These phenomena have had profound consequences for both populations and healthcare systems. The SARS-CoV-2 outbreak and recent extreme climate events are notable examples [2]. Although these crises differ in nature, they often converge toward similar socio-economic, technological, and organizational impacts. For instance, the COVID-19 pandemic led to a global lockdown and revealed the limitations not only of healthcare systems but also of socio-economic infrastructures. In the prehospital care sector, human-induced disasters, natural hazards, mass causality events, and epidemics have exposed several systemic weaknesses, including delays in patient care, extended response times, overwhelmed emergency call centers, resource shortages, increased violence toward healthcare personnel, and overcrowded emergency departments[3,4]. Addressing these challenges requires modernizing workflows through the integration of advanced technological applications to improve the efficiency and effectiveness of EMS, enhance responses to critical situations, and optimize patient outcomes. The strategic position of EMS at the interface between the population and healthcare facilities makes them key actors in developing, coordinating, and planning proactive measures. These include early warning systems, evacuation planning, and optimal resource allocation, all of which are essential to meeting the considerable challenges posed by natural disasters and health crises.

While humanity faces an ever-increasing demand for emergency services—driven not only by health crises and natural disasters but also by population aging and the misuse of emergency services for non-urgent cases, AI tools such as natural language processing, computer vision, and robotics are experiencing rapid advancement. They are transforming multiple domains, including industry, commerce, and, increasingly, healthcare. These technological developments offer promising opportunities to enhance productivity, modernize emergency departments, and strengthen responses to crises. AI also holds great potential to improve disaster management through predictive and recommendation systems capable of analyzing large, heterogeneous datasets while accounting for uncertainty to anticipate risks and forecast potential disasters. In this study, we conduct a literature review to explore how AI’s potential and capabilities can be leveraged to address natural disasters, epidemics, and other large-scale crises. The objective is to identify the progress made in applying AI to improve early detection of epidemic clusters, prediction of natural disasters, forecasting of emergency medical demand, and resource planning, thereby supporting decision-makers in preparedness and response activities. By optimizing operational processes among stakeholders involved in crisis management, this approach aims to strengthen the resilience of healthcare systems and improve outcomes for affected populations. Enhancing responses to health emergencies and natural disasters remains essential for ensuring the stability and safety of modern societies.

In France, although public health institutions such as Santé publique France (Public Health France), the Institut Pasteur (Pasteur Institute), the Réseau Sentinelles (Sentinel Network), and Météo-France (French National Meteorological Service) are actively addressing the challenges of health emergencies and natural disasters, few EMS currently integrate AI-based solutions into their workflows. Yet such tools could be used to forecast emergency medical demand, anticipate surges in activity, or optimize human and material resource allocation. The integration of these technologies within the EMS system remains at an early

stage, mainly due to challenges related to the General Data Protection Regulation (GDPR), organizational resistance to change, lack of interoperability among hospital information systems, and the absence of a clear regulatory framework for clinical validation of algorithms. Conversely, the development and adoption of AI tools in EMS have advanced significantly in several Asian and Anglo-Saxon countries. This trend is driven by greater institutional flexibility, a culture of technological innovation, and historical exposure to major crises. For example, Japan has invested heavily in early warning and crisis management systems following the Fukushima nuclear disaster, while China has accelerated the deployment of disease surveillance and prediction tools in response to the COVID-19 pandemic. Moreover, numerous international collaborations, particularly between the United States and Asian countries illustrate the growing global interest in AI applications for emergency and disaster management.

The remainder of this paper is organized as follows: Section 2 introduces the key conceptual and theoretical elements underpinning this literature review. Section 3 outlines the methodology used for article retrieval and selection, following the PRISMA protocol. Section 4 presents descriptive statistics of the selected publications after applying inclusion and exclusion criteria. In Section 5, we describe the experimental protocol used to identify research themes, which we compare with those generated through LiteRev, an automated literature analysis tool. The results of this comparison are also discussed. Section 6 highlights several AI applications identified in the reviewed studies, categorized into research areas such as: Environmental and Disaster Risk Forecasting, Forecasting Emergency Demand and Managing Prehospital Resources during Epidemic Outbreaks, Prehospital Emergency Triage, Prediction, and Outcome Assessment, and Patient-Centered Prediction. Finally, Section 7 identifies research gaps, methodological limitations, and future research directions, while Section 8 summarizes the key findings and outlines perspectives for integrating AI into EMS for responding to health emergencies and disasters.

2. Background and Foundations

2.1. The role of EMS to mitigate Health Emergencies and Disasters

Emergency Medical Services (EMS) constitute a critical component of healthcare systems due to their proximity to the population and their role in the early management of acute conditions. For many time-sensitive and life-threatening events, the patient care pathway begins with pre-hospital assessment and stabilization by EMS, followed by hospital admission and definitive in-hospital care. Positioned at the interface between the community and healthcare facilities (Figure 1), EMS generate real-time operational and clinical data that can support syndromic surveillance and anomaly detection. This strategic role makes EMS particularly well-suited for contributing to early warning systems aimed at mitigating mass-casualty incidents, health emergencies, and large-scale disasters.

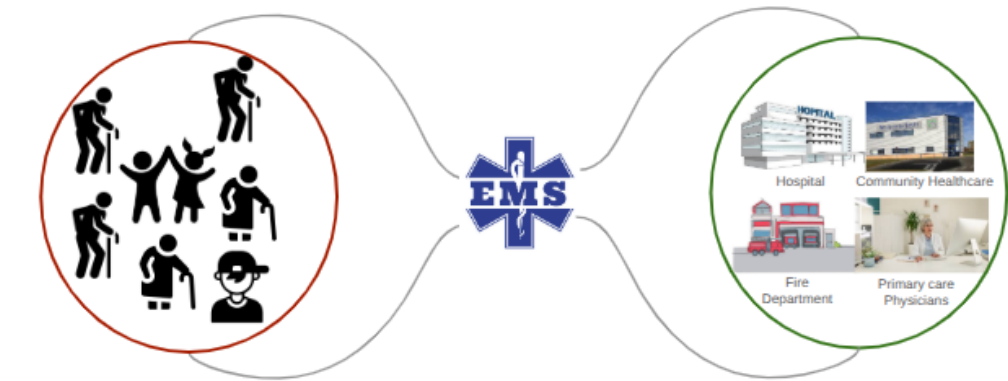


Figure 1. Strategic positioning of EMS in healthcare systems.

Although Emergency Medical Services (EMS) are present in most countries worldwide, the extent of their population coverage and operational capacity is strongly associated with national economic resources, particularly Gross Domestic Product (GDP) per capita, as illustrated in Figure 2. Traditionally, EMS systems have been described according to two main organizational models: the Anglo-Saxon model and the Franco-German model. These models differ in governance structure, level of pre-hospital medicalization, and care delivery philosophy. EMS systems typically include emergency call centers responsible for call triage, medical dispatch, resource allocation, and directing patients to the most appropriate healthcare facilities.

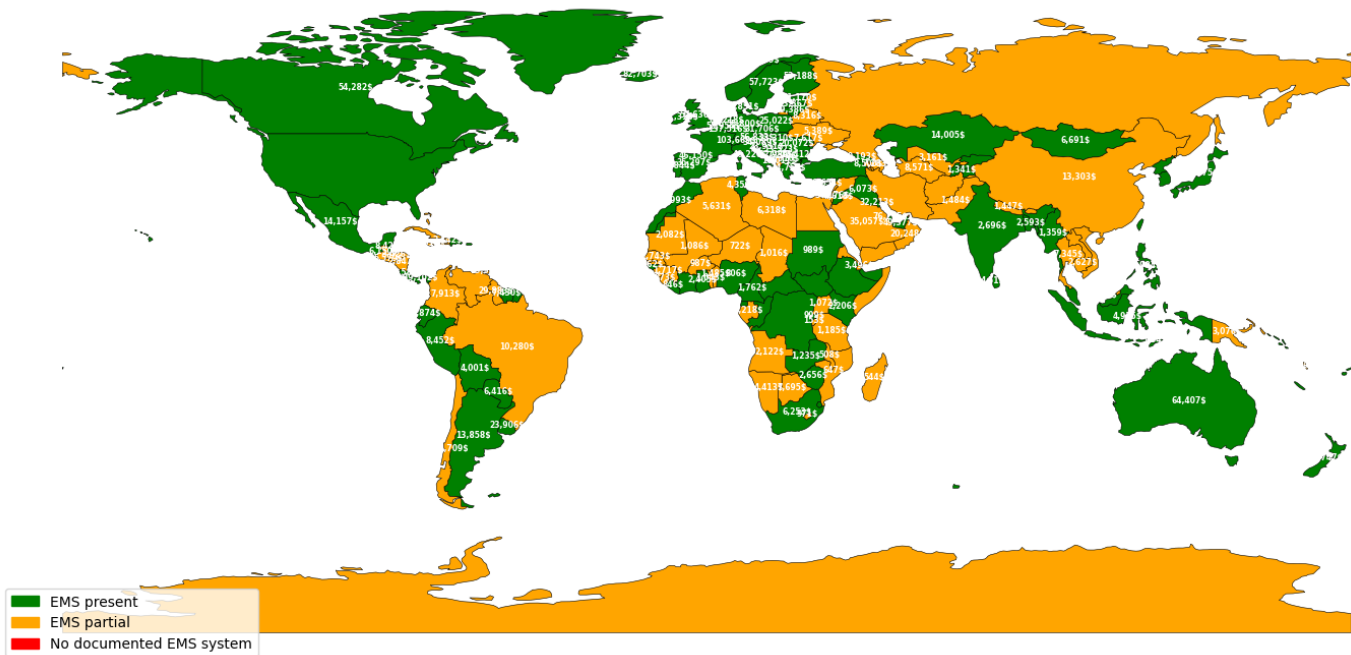


Figure 2. EMS distribution across the world according to mean GDP. This figure shows the availability of emergency medical services (EMS) in each country. Countries in red indicate regions where EMS coverage is limited or partial.

2.2. Overview on Artificial Intelligence

2.2.1. Theoretical Foundations and Taxonomy of Artificial Intelligence

AI refers to a set of computational methods and theoretical frameworks designed to enable machines to perform tasks that typically require human intelligence [5]. These methods allow systems to extract knowledge from structured and unstructured data, supporting reasoning, prediction, decision-making, and automation. The theoretical foundations of AI are deeply rooted in formal logic[6], probability theory[7], information theory[8], optimization theory[9], and algebraic structures[10]. These foundations enable the design of algorithms capable of learning patterns, modeling uncertainty, and solving complex real-world problems.

AI can be categorized into several major paradigms, including Symbolic Artificial Intelligence, Machine Learning (ML), Deep Learning (DL), and Generative AI [11]. Symbolic AI relies on explicit rule-based reasoning and knowledge representation. In contrast, ML focuses on data-driven model construction, allowing systems to learn patterns directly from empirical data. Deep Learning extends ML by leveraging multi-layer neural networks to process high-dimensional inputs [12]. More recently, Generative AI has emerged as a powerful paradigm capable of producing new content, such as text, images, and structured medical reports.

2.2.2. Machine Learning Paradigms

ML enables empirical knowledge extraction from data through structured learning processes. Depending on data characteristics and task objectives, different learning paradigms are employed: supervised learning, unsupervised learning, semi-supervised learning, and reinforcement learning[14,15].

The choice of paradigm depends on whether the data are labeled or unlabeled, static or sequential, and whether the objective is prediction, pattern discovery, or decision optimization. These paradigms form the methodological backbone of AI applications in healthcare and Emergency Medical Services (EMS).

2.2.3. Deep Learning for High-Dimensional Data

ML that utilizes artificial neural networks with multiple hidden layers to model complex representations [16]. These models are particularly effective when dealing with high-dimensional data such as medical images, audio recordings of emergency calls, electrocardiogram signals, and free-text clinical notes.

High dimensionality often introduces computational challenges and reduces distance-based discrimination between observations. Convolutional Neural Networks (CNNs), for instance, address this issue by applying convolutional operations to extract invariant and hierarchical features from images, thereby reducing computational burden while preserving relevant clinical information. In EMS applications, deep learning models have been used for medical image interpretation, voice-based triage systems, and automated clinical documentation support.

2.2.4. Supervised, Unsupervised, and Semi-Supervised Learning Approaches and Generative Artificial Intelligence

In supervised learning, models are trained using labeled datasets, where each input is associated with a known output. The objective is to learn a mapping function capable of performing classification or regression tasks on unseen data [17]. Common supervised algorithms include Decision Trees, Ensemble Learning methods (e.g., Random Forests and Gradient Boosting), Support Vector Machines (SVM), and Logistic Regression. In the EMS context, supervised learning has been widely applied to patient-centered outcome prediction, classification of emergency call priorities (e.g., high-priority P0 calls versus moderate-priority P2–P3 calls), severity estimation, and EMS demand forecasting [18]. Accurate predictive modeling contributes to improved triage accuracy, optimized dispatching decisions, and enhanced allocation of emergency resources.

Unsupervised learning focuses on uncovering hidden structures within unlabeled data [19]. Unlike supervised learning, no predefined target variable is provided. Instead, algorithms identify intrinsic patterns, clusters, or latent representations. Common unsupervised methods include K-means clustering, hierarchical clustering, and dimensionality reduction techniques such as Principal Component Analysis (PCA). These approaches are particularly useful in healthcare settings where large volumes of unlabeled data are available. For example, clustering techniques can help identify patient phenotypes or patterns of EMS utilization [20]. In deep learning, autoencoders constitute a major unsupervised architecture. They project input data into a lower-dimensional latent space and reconstruct it to ensure that the extracted features retain meaningful variance. Such representation learning techniques help address the curse of dimensionality while enabling efficient data compression and anomaly detection.

Semi-supervised learning combines elements of supervised and unsupervised learning to leverage partially labeled datasets [21]. In many healthcare applications, labeling data is costly and time-consuming. Semi-supervised approaches use unlabeled data to infer

pseudo-labels or extract informative features before applying supervised training. This hybrid paradigm has contributed significantly to the emergence of Generative Artificial Intelligence, particularly through Transformer-based architectures [22]. Transformers rely on attention mechanisms that model contextual dependencies within sequential data, revolutionizing Natural Language Processing (NLP) and related domains. State-of-the-art generative models such as *GPT*, *LLaMA*, and *Bard* are built upon Transformer architectures and have demonstrated strong performance in tasks such as clinical document generation from doctor–patient conversations, medical speech summarization, and clinical question answering. These systems have contributed to reducing error rates and improving contextual coherence in automated healthcare documentation.

2.2.5. Reinforcement Learning for Sequential Decision-Making

Reinforcement Learning (RL) addresses sequential decision-making problems by allowing an agent to interact with an environment through trial and error[23]. The agent selects actions, receives rewards or penalties, and iteratively improves its policy to maximize cumulative rewards.

Common RL algorithms include Q-learning, Deep Q-learning, and Actor–Critic methods. Depending on whether an internal model of the environment is used, RL approaches are categorized as model-based or model-free [24]. Model-based RL relies on a representation of environmental dynamics to support planning, whereas model-free RL learns optimal policies directly from observed interactions.

In healthcare and EMS systems, RL has demonstrated strong effectiveness in optimizing ambulance dispatching, dynamic resource allocation, and personalized treatment strategies. Recently, Reinforcement Learning from Human Feedback (RLHF) has been integrated with Large Language Models to align model outputs with human preferences and enhance decision quality in complex clinical and operational scenarios.

2.2.6. AI Applications in Emergency Medical Services

AI applications have the potential to significantly transform EMS practices, particularly in the context of disasters and health emergencies. Predictive models can anticipate surges in emergency calls and support proactive resource deployment, which is critical during catastrophic events or pandemic outbreaks. Forecasting tools enable optimized ambulance routing and reduced response times, directly impacting patient outcomes.

AI-driven triage systems enhance prioritization accuracy by extracting relevant clinical features from call-center conversations and patient assessments. Furthermore, intelligent surveillance systems act as sentinel mechanisms capable of detecting abnormal patterns in EMS demand, thereby strengthening preparedness for disasters and large-scale health emergencies.

By optimizing dispatch strategies, supporting diagnostic decision-making, and enhancing situational awareness, AI contributes to increased operational efficiency, reduced costs, and improved quality of pre-hospital care delivery. Recent studies systematically reviewed indicate that these AI-driven interventions can significantly improve EMS responsiveness under conditions of extreme demand and uncertainty.

2.3. Overview on Uncertainty in AI

Accounting for uncertainty is critical and paramount in high-uncertainty sectors such as Emergency Medical Services (EMS). It represents a key avenue for ensuring reliability, security, transparency, and trust in AI and ML systems. Uncertainty awareness significantly influences decision-making by shaping choices, confidence, and caution. Recognizing that any forecast, prediction, or recommendation generated by an AI system is associated with

a certain degree of uncertainty allows the end-user to adopt a more critical and reflective attitude toward the produced results.

AI and ML systems that are capable of quantifying uncertainty and communicating it to human users along with explanations of its underlying sources—are referred to as uncertainty-aware AI/ML systems [26]. The uncertainty-related knowledge transferred from the system to the human operator acts as a regulating factor in the decision-making process. When uncertainty is high, it introduces legitimate doubt that may extend the time required to reach a decision, thereby encouraging a more thorough analysis of the available information. Conversely, low uncertainty can enhance confidence and accelerate decision-making.

One of the main objectives in the development of AI systems is to reduce the specific uncertainties associated with the problem for which the system is designed [27]. However, it is evident that AI itself becomes subject to various forms of uncertainty, generating additional forms such as epistemic uncertainty, actionable uncertainty, and ethical uncertainty [28]. These forms of uncertainty arise from multiple sources, including data uncertainty, computational uncertainty, model uncertainty, and environmental uncertainty — and are inherently linked both to the methods and tools employed in AI research and development and to the way these methods are applied.

They are generally classified into two main categories: epistemic uncertainty and aleatoric uncertainty [29,30]. While aleatoric uncertainty stems from stochastic perturbations in the data-generating process, epistemic uncertainty arises from incomplete knowledge of the model parameters, structure, or of its representativeness with respect to the system it aims to mimic [31]. Leveraging information theory, probability theory, and statistical tools such as entropy, information gain, probability density functions, cumulative distribution functions, and confidence intervals is essential for uncertainty quantification in order to make more informed decisions in highly stochastic environments.



Figure 3. Mind map on uncertainty and the methods and tools for its quantification and evaluation.

3. Material and Methods

3.1. Literature search strategy

The systematic review focuses on a central question: What AI methods and tools have been developed and applied in EMS to address disasters and health emergencies? To answer the research question, several online databases were consulted, including PubMed, Web of Science, Scopus, IEEE Xplore and Embase. The research databases were selected based on their specificity and relevance to the application of AI in medical research. While PubMed and Web of Science are widely recognized as leading references in the healthcare sector and enable access to a broad range of biomedical and clinical studies, Web of Science and IEEE Xplore provide extensive coverage of empirical studies and quantitative methodologies. In contrast, Embase offers a complementary perspective, particularly on studies focusing on the application of AI in healthcare. Consequently, Embase was used in the final stage to compensate for the imbalances observed in the topics covered by PubMed, IEEE Xplore, and Web of Science.

3.2. Search terms

Given their strategic role at the interface between the population, healthcare systems, and public authorities, EMS have been the focus of studies investigating the use of AI to enhance early warning, monitoring, and emergency response planning. To capture this evidence, the literature search was structured into two complementary keyword sets. The first set covers studies addressing the prehospital context, including emergency medical regulation, public health surveillance, demand forecasting, resource allocation and deployment, and initial triage procedures.

Table 1. Search terms: AI in the prehospital setting.

Themes	Search Terms
Artificial Intelligence	"Artificial Intelligence", "AI", "Machine Learning", "Deep Learning", "Natural Language Processing", "Large Language Models", "GPT", "BERT", "LLM"
Prehospital Care	"Prehospital", "Emergency Medical Services", "EMS", "Out-of-Hospital", "Paramedic", "Ambulance", "Emergency Call"
Triage and Dispatch	"Triage", "Dispatch", "Emergency Dispatch", "Triage System", "Prioritization", "Severity Assessment"
Prehospital Clinical Tasks	"Diagnosis", "Risk Assessment", "Clinical Decision Support", "Care Pathway", "Patient Stratification", "Forecasting"
Data Modalities	"Audio Data", "Free Text", "Clinical Notes", "Dispatch Text", "ECG", "Time Series Data", "Multimodal Data"
Clinical Outcomes and Performance	"Mortality", "ICU Admission", "Hospital Admission", "Response Time", "Time to Treatment"
Crisis / Public Health Emergency Context	"Health Emergencies", "Early Warning Systems", "Mass Casualty Incidents"[MeSH], "Disasters"[MeSH], "Major Event"

The second set focuses on AI use within hospital settings, from emergency department admission to hospitalization and critical resource management. Harmonizing these two components reflects the operational logic of public health emergencies, which unfold across multiple levels of care. It also enables a comprehensive assessment of AI contributions along the entire prehospital–hospital care pathway, in alignment with the “Patients–Resources–Territories” meta-model.

Table 2. Search terms: AI in the intrahospital setting.

Themes	Search Terms
Artificial Intelligence	"Artificial Intelligence", "AI", "Machine Learning", "Deep Learning", "Neural Networks", "Explainable AI (XAI)", "Uncertainty Quantification"
Emergency Care & Triage	"Emergency Department", "ED Triage", "Hospital Triage", "Severity Classification"
Clinical Prediction	"Clinical Deterioration", "Patient Outcome Prediction", "Risk Stratification", "Critical Care Need"
Flow and Resource Management	"Hospital Demand Forecasting", "Bed Management", "Resource Allocation", "Crowding Prediction"
Data Modalities	"Vital Signs", "Electronic Health Records", "Clinical Notes", "Time Series Data", "Multimodal Data"
Clinical Outcomes and Performance	"ICU Admission", "Hospital Admission", "Length of Stay", "Clinical Outcomes"
Crisis / Public Health Emergency Context	"Health Emergencies", "Disaster Medicine", "Mass Casualty Incidents"

3.3. Inclusion and Exclusion Criteria

The inclusion and exclusion criteria were defined and then validated by peers. The research team included physicians to ensure the relevance and quality of the various topics explored. Only studies published between 2015 and July 2025 were included. On Scopus, a date filter was automatically applied, as the initial results were deemed too extensive, making a proper review infeasible. By deliberate choice of the author, the search did not include non-indexed databases or "gray literature", in order to prioritize validated and easily traceable sources.

Before detailing the reasons for the inclusion or exclusion of articles during the "screening" phase, it was required that articles be unique in the database. Articles that had been retracted were systematically excluded. Studies were included if their titles contained the predefined keywords listed in Tables 1 and 2. Additionally, articles with titles framed as open-ended questions, those focused exclusively on AI without direct relevance to emergency systems, as well as narrative or systematic reviews, were excluded. Therefore, the articles included in this study had to be empirical studies, propose new quantitative methods, or demonstrate the applicability of AI in emergency management for crises and disasters through case studies, empirical evidence, or scenarios.

The exclusion criteria applied after reading the abstracts and conducting a detailed analysis of the full texts primarily relied on a lack of scientific rigor in the AI methods proposed, or the superficial nature of the approaches presented. Articles that mentioned AI without proposing a new method or clearly specifying the type of AI used to address the studied problem were also excluded.

Regarding articles focused on forecasting, prediction, or estimation, particular attention was paid to the presence of evaluation methods for the proposed models, as well as explainability mechanisms. Only articles presenting performance results and meeting the explainability requirements were considered relevant. Furthermore, the articles had to present an empirical research methodology ensuring the reproducibility and transparency of the studies, a criterion that also led to the exclusion of some works. Finally, the following were also excluded:

- Reason 1 : Redundant articles, where a more relevant and similar article had already been included;

- Reason 2 : Articles using exclusively hospital data or synthetic data, or those that focused solely on hospital-related issues.
- Reason 3 : Other issues, for example, the study is not relevant after deep screening.

The review also included relevant studies identified beforehand, which we refer to as the "Article Radar" method. In addition, the "snowball" method, which involves exploring the references of studies deemed highly relevant, was used to discover more studies, while applying the inclusion and exclusion criteria.

3.4. Article selection process

The documentation of the article selection process following the PRISMA protocol is summarized in the figure below. The flow diagram is structured into three main phases: identification, screening, and inclusion. The top-down approach proposed by the PRISMA methodology enables progressive filtering while indicating, at the end of the process, the total number of articles retained for inclusion in the systematic review, as shown in the diagram below (138 articles). This total number results from the sum of articles included in the previous study (66 articles) and those derived from the new study.

For the new study, querying the different databases using the predefined search equation led to the identification of a total of 2,531 articles. A specific method was then applied to remove duplicates, resulting in the elimination of 234 redundant articles. Additionally, 8 retracted articles were excluded. These steps resulted in 2,239 articles being retained for the next phase. During the title and abstract review, 2,204 articles were deemed irrelevant and excluded, leaving 85 studies for full-text evaluation. Following the full-text review, and based on the predefined inclusion and exclusion criteria, 67 articles were selected for inclusion.

To balance and enrich the corpus of studies, the Radar method and targeted identification of articles in the Embase database were used, allowing for the identification of 5 additional articles, bringing the total number of articles from the new study to 72.

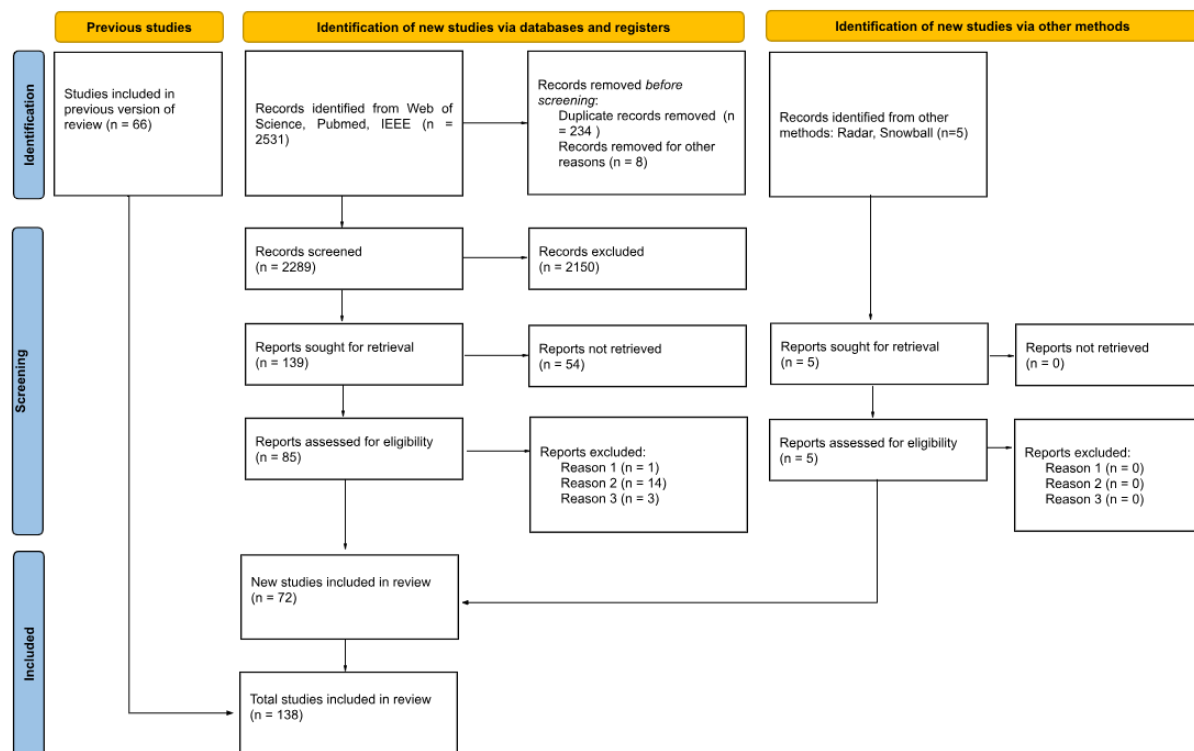


Figure 4. Flow chart - Prisma

4. Descriptive statistics of selected papers

This section uses multiple descriptive analytic approaches, including frequency measures, central tendency, dispersion, and positional metrics, to describe recurring patterns and extract relevant insights from the selected papers.

4.1. Number of publications over time

The selected articles exhibit a temporal structure, characterized by an uneven distribution over time. Figure 5 presents the percentage distribution of publications per year, illustrating the evolution of research activity on the application of AI in EMS for health emergencies and disasters.

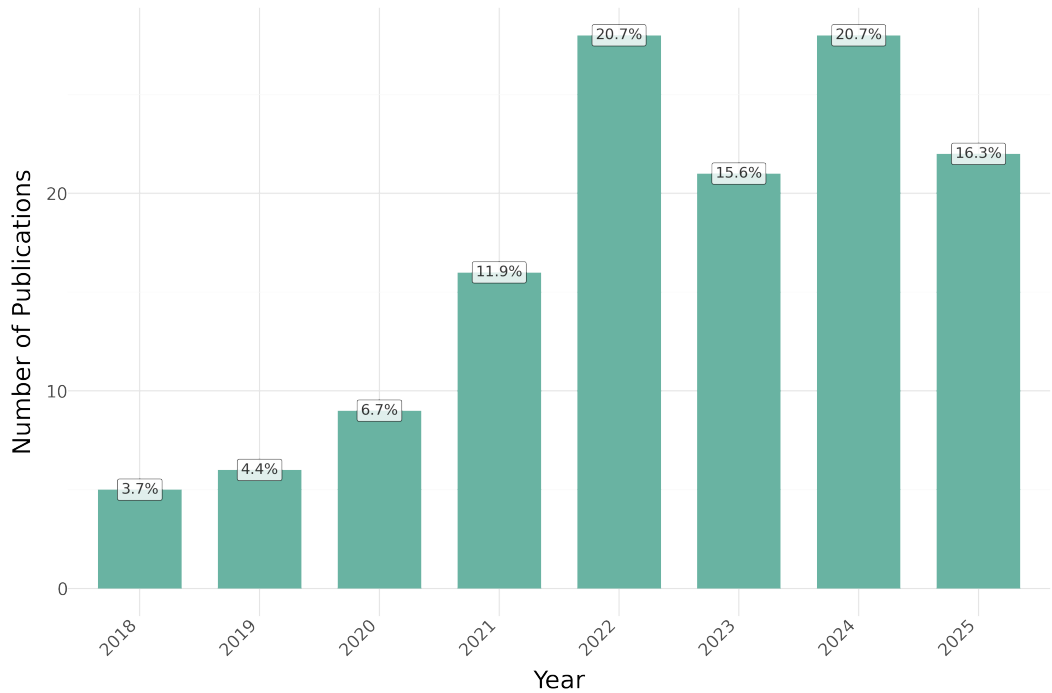


Figure 5. Percentage distribution of publications over the years.

The results indicate a significant rise in the quantity of publications published in recent years, demonstrating an increasing interest among researchers in this area of study. This trend also illustrates the field’s progression toward greater scientific maturity: the research is progressively getting more fruitful, indicating that prior findings and past mistakes are being integrated into future studies. The recent increase in publications can be attributed also to the recurring occurrence of emergencies that place significant pressure on healthcare systems and public authorities. This trend reflects the substantial involvement of both researchers and public institutions in the development and validation of projects aimed at advancing artificial intelligence methods to enhance public health emergency preparedness, optimize resource allocation, and improve response effectiveness.

4.2. Most used words in the titles

The titles of the articles selected using the PRISMA methodology reveal the terminologies most frequently employed by the authors. These terms represent key concepts summarizing the various solutions proposed to address the issues discussed in the articles. They also support the validity of the publication selection, reflecting the emergence of the terms that guided the article choices, and serve as a framework for identifying the main research topics. In our study, relevant recurring terms include forecasting, machine learning, prediction, language models, prehospital care, emergency medical services, demand, triage,

and others (see Figure 6). These terms help situate our study within a specific context defined by this set of key concepts.



Figure 6. Most frequent words appearing in article titles.

4.3. Publications distributions across country

We used a bubble plot to statistically represent the number of publications per country. Only the country of origin of the first author was considered in this study, although an analysis of all author affiliations indicates a strong international interest in the research field. The uneven distribution of publications may be explained by several factors, including research interest, allocated funding, and the global ranking of universities in these countries, which can serve as an additional motivation to excel and produce innovative work.

Our analysis shows that developed countries (e.g., France, the United States, China, Japan, and Australia) have been the most innovative in applying AI methods in EMS to predict natural disasters, mass incidents, and epidemics, addressing both demand forecasting and disaster and epidemic management. This predominance can be explained by the fact that these countries have been directly confronted with life-threatening challenges, such as pandemics and natural disasters. For example, China was at the epicenter of the COVID-19 pandemic [32], and Japan faces environmental challenges [33], including air quality issues, which require advanced technological tools for monitoring.

Similarly, the distribution of publications, as visualized through bubble plots, aligns closely with the structure of existing prehospital care systems, particularly the Pan-Asian [34], Anglo-Saxon [35], and Franco-German systems [36]. This correlation underscores the absence of comparable systems in less developed countries, which may contribute to their lower scientific output in this field.

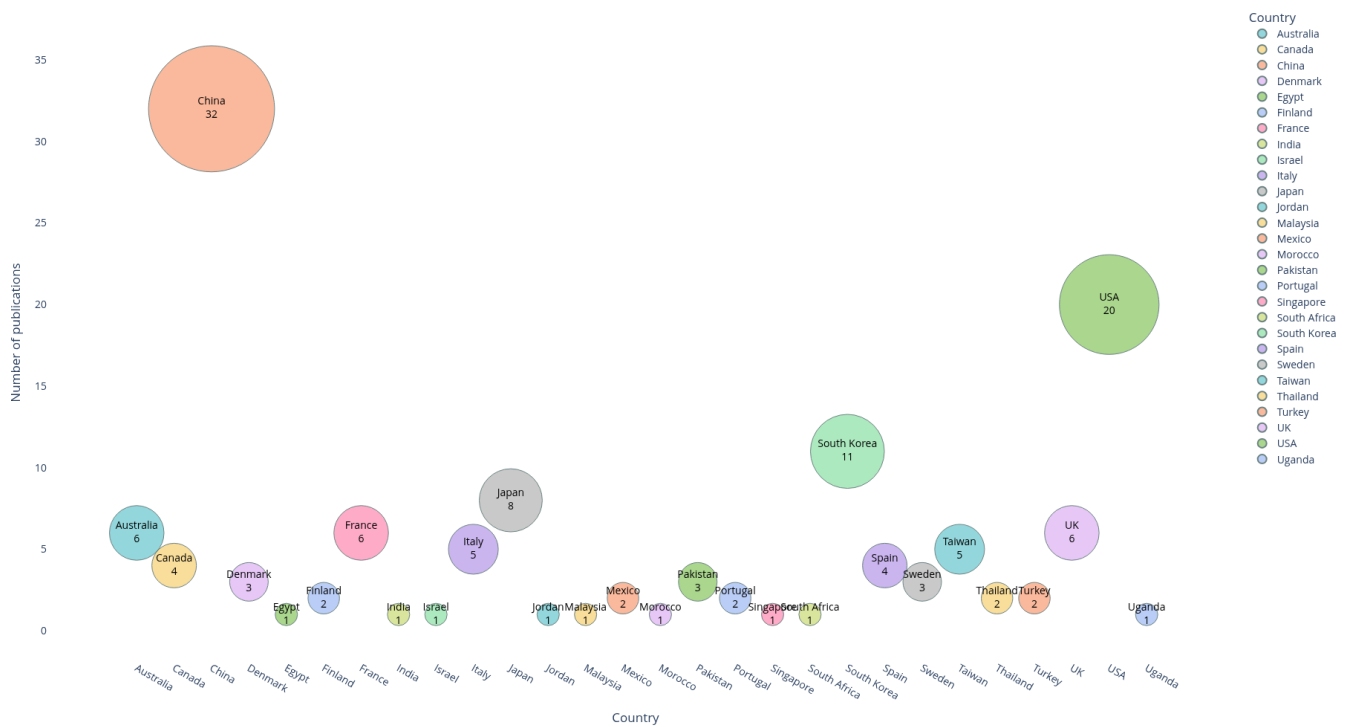


Figure 7. Number of publications by country.

5. External Validation of Initially Selected Thematics

In this section, we present a methodology for the external validation of study thematics initially defined subjectively. The proposed approach consists of two steps: first, the articles are grouped using unsupervised learning techniques to position the initial themes relative to the obtained clusters; second, these themes are adjusted using the results provided by the automated literature analysis tool LiteRev [37]. The objective is to identify “natural” themes within the corpus and to validate or adjust the initially defined themes, thereby minimizing human bias in their selection.

5.1. Experimental Protocol

We hypothesize that analyzing titles, abstracts, and keywords enables the identification of the key themes addressed in the Discussion section of the literature review. In our experiment, we use the titles and abstracts of the selected articles, which are projected into a high-dimensional vector space using the pre-trained *all-mpnet-base-v2* model. This model leverages advanced semantic embedding techniques and has been shown to outperform traditional static word embedding models such as *Word2Vec* and *GloVe* in semantic similarity tasks. To minimize the influence of low-information terms and enhance the quality of the vector representations, we apply dedicated linguistic preprocessing steps.

The curse of dimensionality, which arises from the reduced discriminative power of distance metrics in high-dimensional vector spaces, has been taken into account. In such settings, clustering performance may deteriorate, leading to potential misclassification. To mitigate this issue, dimensionality reduction was performed prior to clustering. In addition, an in-depth analysis was conducted on the articles obtained for each cluster to position the initially defined themes. Based on the cluster analysis, the study themes were refined. Finally, LiteRev was utilized in the final stage to further refine the study themes through manual comparison. The key steps of the experimental process and the tools used are enumerated below:

1.

Dimensionality reduction: application of UMAP (Uniform Manifold Approximation and Projection);

423
2.

Clustering: use of the K-Means++ algorithm to group articles based on their thematic similarities;

424
3.

Data: titles and abstracts extracted from databases such as IEEE Xplore, Web of Science, and Embase, filtered according to PRISMA guidelines to retain only relevant studies;

425
4.

Development environment: implementation in Python 3.11.13.

426

5.2. outcomes

The results of the experimental protocol are presented in Figure 8, which consists of two subfigures: Figures 8a and 8b. These subfigures illustrate the projection of texts, combining the titles and abstracts of the selected articles into a two-dimensional latent space. Figure 8b shows a grouped view of the articles, whereas Figure 8a, which uses the traditional method without dimensionality reduction, displays the articles scattered across the space.

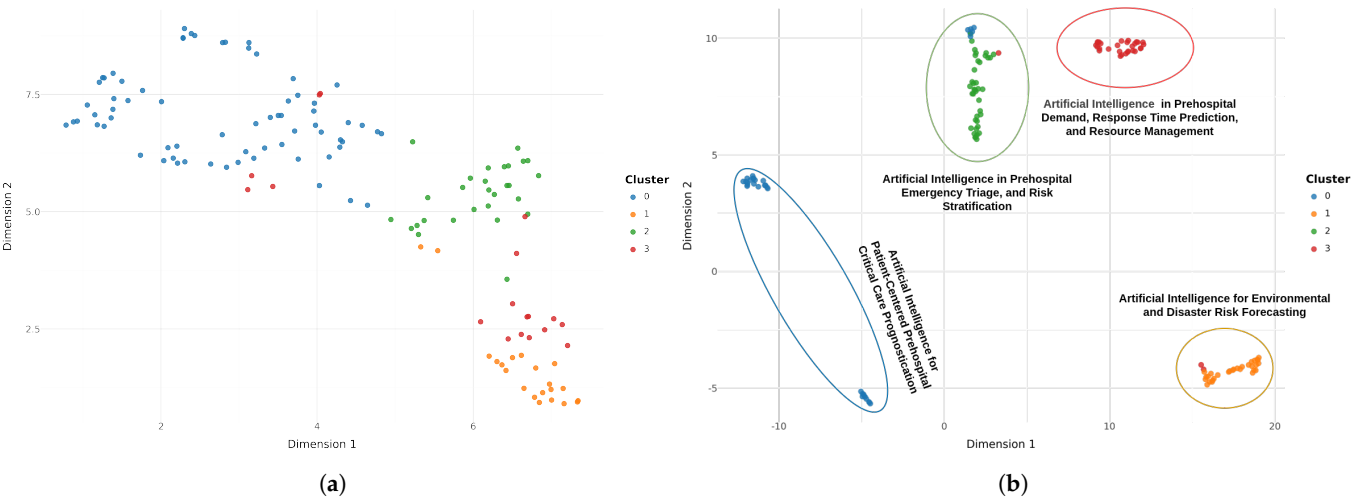


Figure 8. Multidimensional representation of articles stratified across several clusters: (a) shows the clusters inferred directly from the embedding without prior dimensionality reduction. (b) illustrates the clusters obtained after applying dimensionality reduction. The corresponding thematic topics associated with each cluster are also indicated.

The number of clusters, which was determined to be four for both subfigures using the elbow method, is the same in both cases. The less cohesive grouping observed in Figure 8a is due to the classical clustering approach, which does not address the curse of dimensionality. Based on the analysis of the titles and abstracts within each cluster, the research themes presented in Table 3 were defined.

However, it is important to note that no specific cluster emerges for the prediction of epidemic diseases, seasonal influenza-like illnesses, or acute respiratory diseases. This may be because the articles addressing these problems primarily focus on resource allocation. In fact, forecasting respiratory diseases serves to guide optimal resource distribution while reducing uncertainty regarding demand, as highlighted by [38]. Consequently, demand forecasting and the prediction of epidemic diseases, seasonal influenza-like illnesses, and acute respiratory diseases are interrelated issues.

The results obtained with the LiteRev software further validate our selected research topics. The clusters generated by LiteRev are shown in Figure 5, where each point represents a selected article. LiteRev identified nine clusters: eight of these correspond to the research

themes presented in Table 4, while the ninth contains articles that could not be assigned to any cluster. Together, these results underscore the relevance of addressing the following themes:

- Artificial Intelligence for environmental and disaster risk forecasting
- Artificial Intelligence for prehospital demand and response time forecasting and resource management ;
- Artificial Intelligence in Prehospital Emergency Triage, Risk Stratification ;
- Artificial Intelligence for patient-centered prehospital critical care prognostication.

Table 3. Thematic clusters identified from titles and abstracts.

Cluster	Main Topic(s)	Number of Articles
0	Artificial Intelligence for patient-centered prehospital critical care prognostication	34
1	Artificial Intelligence for environmental and disaster risk forecasting	30
2	Artificial Intelligence in Prehospital Emergency Triage and Risk Stratification	39
3	Artificial Intelligence in Prehospital Demand Forecasting, Response Time Prediction, and Resource Management	32

Table 4. LiteRev-Based Identification of Thematic Clusters.

Cluster	Main Topic(s)	Number of Articles
1	Flood, model, disaster, dataset, datum, prediction, feature, propose, base, time	186
2	Patient, care, medical, emergency, clinical, hospital, triage, health, system, service	178
3	Disease, outbreak, health, population, mortality, death, country, public, transmission, virus	160
4	fire, model, water, coal, emission, area, quality, concentration, datum, air	154
5	health, public, pandemic, emergency, infectious_disease, healthcare, response, datum, surveillance, system	145
6	climate, model, forecast, change, dengue, extreme, rainfall, temperature, weather, precipitation	133
7	network, resilience, algorithm, disaster, evacuation, system, datum, propose, power, base	125
8	AI, disaster, research, emergency, technology, artificial_intelligence_ai, potential, management, care, system	73
-	unclassified papers	232

6. Results

This section presents the results and key findings from the selected articles. It is organized around our primary research topics, which were refined through an article clustering strategy and an external validation stage facilitated by the LiteRev software. Throughout the section, we highlight cutting-edge approaches proposed in the literature, along with key insights from pioneering studies on health emergencies. The section is structured into multiple parts: one focusing on AI applications for environmental and disaster risk monitoring, and another concentrating on the use of AI in Emergency Medical Services (EMS) for managing health emergencies, specifically at the resource allocation stage, patient triage, and clinical outcomes prediction.

6.1. Artificial Intelligence for Environmental and Disaster Risk Forecasting

6.1.1. Artificial Intelligence for Forecasting Environmental Factors

Environmental factors are key contributors to the development of many epidemiological diseases. For instance, previous studies have shown that environmental exposures play a significant role in the etiology of respiratory diseases such as asthma, as well as cardiovascular diseases [39,40]. Environmental determinants therefore represent a major

public health challenge, not only because of the substantial risks they pose to populations, but also due to their cascading effects across different stages of the life course [45]. More precisely, exposure to water, air, and soil pollution and substandard housing conditions contributes to the development of numerous diseases, including cancers and reproductive and developmental disorders. It has been demonstrated that certain diseases may be triggered or exacerbated by environmental factors, highlighting the need to develop effective tools for monitoring environmental agents. As highlighted by [44], the use of AI for extracting knowledge from air pollution data and modeling past feedback (RETEX) on the spatiotemporal dynamics of air pollutants is essential for effective surveillance and the prevention of emerging risks.

Wang et al. [42] proposed a cutting-edge framework to extract spatiotemporal features for prediction purposes from air quality data collected by monitoring stations in Beijing (China). The spatiotemporal properties were derived through advanced signal processing techniques, enabling the extraction of Intrinsic Mode Functions (IMFs). Specifically, the PM_{2.5} time series was decomposed into K sub-sequences using the Variational Mode Decomposition (VMD) method, with each component capturing a distinct frequency-centered characteristic of the original signal. The decomposed components were then spatially represented and processed through a Graph Attention Network (GAT) to learn cross-feature spatial dependencies. In the final stage, a sequential modeling approach was applied to the extracted spatial representations in order to capture temporal dynamics and build the forecasting model. Similarly, [46] addressed the same problem by proposing an alternative methodology. Instead of modeling the time series using the Bi-LSTM algorithm, they proposed a stacking model that combines a sequential data modeling algorithm, namely LSTM, with a decision-tree-based algorithm called Random Forest. By doing so, the authors aimed to increase the model's complexity and predictive capability, ultimately achieving superior performance while maintaining interpretability of the overall architecture.

Wind speed is a key meteorological variable influencing the dispersion and transport of air pollutants and is also associated with an increased risk of wind-induced accidents and extreme events. Consequently, accurate wind speed forecasting is essential for informed decision-making in the healthcare sector and public institutions. Xuhui et al. [47] proposed a hybrid framework combining Long Short-Term Memory (LSTM) networks and Gaussian Mixture Models (GMM) for short-term forecasting of mean wind speed and turbulence intensity. However, the chaotic nature inherent in the behavior of meteorological elements complicates their accurate prediction and control. To mitigate prediction errors, Ruibo Qiu et al. [50] employed more robust time series models based on transformer architecture, as opposed to RNN-based models, which, despite their fast training, did not yield optimal results. By comparing the Transformer with its variants, the study suggests that transformer-based algorithms achieved the lowest loss score, although the Transformer itself also performed well on the task. The findings of [51] highlight the significant impact of weather elements such as wind patterns, air dynamics, and temperature on human health. Furthermore, when extreme environmental conditions are met, they can lead to catastrophic events such as forest fires, extreme temperatures, and severe weather disasters, which are key contributors to drought. To address these challenges, the authors proposed a deep learning algorithm that extracts invariant features of weather elements, represented as a 1D vector. These invariant characteristics, which capture local patterns, are subsequently modeled using an attention-based Gated Recurrent Unit (GRU) network to capture cross-relations and temporal dependencies, with the aim of improving the accuracy of weather forecasting.

Table 5. Thematics related to the use of Artificial Intelligence for environmental and disaster risk forecasting.

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[41]	Forecasting global-scale temperature and pressure	Meteorological data	-	ML/DL (Gaussian Processes, ANN)	Yes	Yes	Yes	UK
[42]	Forecasting air-quality (PM _{2.5}) concentration) using a hybrid spatio-temporal DL model	Air quality data	PM ₁₀ , CO, O ₃ , NO ₂ , Meteorological variables	ML/DL (Hybrid model: VMD-GAT-BiLSTM)	No	No	No	China
[43]	Classifying natural disasters through images	Natural disaster images	-	DL (VGGNet19, ResNet50, ViT)	No	Yes	No	South Korea
[44]	Prediction of air pollution using piecewise affine models	Air quality and meteorological data	PM _{2.5} , PM ₁₀ , CO, O ₃ , NO ₂ , Temperature, Humidity, Wind speed	ML (Piecewise Affine Regression Models)	No	No	Yes	China
[45]	Predicting PM _{2.5} concentration	Beijing air quality data	Spatiotemporal features, PM ₁₀ , CO, O ₃ , NO ₂ , SO ₂ , Temperature, Pressure, Wind speed, Wind Direction, Precipitation, Dew Point	ML/DL (1D CNN, BiLSTM, Attention mechanisms)	No	Yes	No	China
[46]	Predicting PM _{2.5} concentration in 7-day window	Air pollution, meteorological data, and AOD (Aerosol Optical Depth)	PM ₁₀ , CO, O ₃ , NO ₂ , SO ₂ , AQI, EH, AOD, Temperature, Pressure, Wind speed	ML/DL (RF-LSTM-Stacking model)	No	Partial	No	China

6.1.2. Forecasting of Natural and Human-Induced Disasters and Risk Management

Natural and human-induced disasters pose substantial risks to both populations and healthcare systems, as their occurrence can result in multifaceted damage, including demographic, economic, and environmental consequences [52]. Among these impacts, the most severe remains the threat to human life. Unlike infrastructure or material assets, which can be rebuilt or replaced, the loss of human life is irreversible. Consequently, improving the short- and long-term prediction of such disasters is of paramount importance [53]. In this context, AI methods offer promising opportunities for enhancing disaster forecasting and risk mitigation. Early warning systems derived from predictive AI models can provide significant benefits for populations, healthcare institutions, and governmental decision-makers by enabling proactive planning, optimized resource allocation, and improved emergency preparedness.

For instance, [54] addressed the problem of earthquake prediction using a novel methodological framework. To overcome challenges related to data privacy, large-scale heterogeneous datasets, communication delays, transmission latency, and limited processing capacity, the authors adopted a Federated Learning (FL) approach. This technique allows multiple local models to be trained on-site using decentralized data, which are then transmitted through a network to a central FL server for aggregation. The aggregation process is performed using a meta-learning algorithm referred to as the FedQuake algorithm. To further enhance predictive performance, a meta-classifier was trained on the aggregated global model outputs. The study demonstrates that the proposed approach achieves high accuracy and reliability, suggesting its potential integration into earthquake early warning systems.

A particularly meaningful task for EMS is casualty prediction. The ability to estimate the potential number of casualties resulting from natural hazards or human-induced disasters provides valuable information for risk assessment and emergency management planning. To address this challenge, [65] proposed a hybrid machine learning framework that combines both regression and classification tasks to predict the number of casualties during emergency events. Regarding wildfire smoke detection, [55] proposed a customized version of the YOLOv8 model for early wildfire identification. The model is designed to detect smoke in surveillance imagery and generate optimized bounding boxes around the identified regions of interest, indicating the potential onset of a forest fire.

Among natural disasters, biological hazards such as viruses and bacteria can also generate situations in which healthcare demand exceeds existing system capacity. Seasonal respiratory diseases, in particular, represent recurrent stressors for Emergency Medical Services (EMS), as they are associated with substantial morbidity and mortality and can significantly strain EMS each year. To address this challenge, [75] introduced a probabilistic early warning system based on a deep learning architecture known as the Dense Residual Network (Dense ResNet) to predict seasonal influenza-like illness (ILI) positivity rates. The proposed model enables early detection of influenza outbreaks and provides continuous risk estimation signals, thereby supporting proactive decision-making. By improving forecasting accuracy and quantifying uncertainty, the system contributes to reducing operational pressure on EMS services and optimizing the allocation of EMS resources.

Table 6. Thematics related to the use of Artificial Intelligence for environmental and disaster risk forecasting (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[47]	Short-term probabilistic forecasting of wind speed	Wind speed data	-	ML/DL (Gaussian Mixture Model, LSTM)	Yes	No	No	China
[48]	Short-term forecasting of precipitation	Historical precipitation data	-	ML/DL (CEEMDAN + GRU)	No	No	No	China
[49]	Short-term, medium, and long-term forecasting of water quality parameters	Water quality indicators (WT, pH, DO, Turbidity, EC, CODMn, NH ₃ -N)	-	DL (LSTM-Seq2Seq)	No	Partial	No	China
[50]	Forecasting meteorological elements	Temperature (T), Surface Solar Radiation (SSR), Soil Moisture (SM), Surface Latent Heat Flux (ET)	-	DL	No	No	No	China
[51]	Short-term weather prediction	meteorological data	Temporal features, geographic location	DL(1DCNN, Attention, GRU)	No	Yes	No	India
[54]	Predicting earthquake using federated learning	Seismic, Climate, Reservoir data	-	ML/DL(Federated learning: FedQuake)	No	No	No	Pakistan
[52]	Short-term forecasting of mountain flood events and their uncertainty	Hydrological, meteorological data	temperature, humidity, weather, wind speed, etc.	Bayesian DL (Bayesian LSTM, Bayesian Neural Networks)	Yes	No	No	China

Table 7. Thematics related to the use of Artificial Intelligence for environmental and disaster risk forecasting (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[53]	natural and human-induced disasters prediction and risk management	Disaster records, environmental and meteorological data, geospatial data, EMDAT	-	ML/DL(Neural-XGBoost)	No	Yes	No	Thailand
[55]	Wildfire smoke detection for early warning	UAVs images or videos of wildfire-prone areas	-	DL(YOLOv8)	No	No	No	South Korea
[56]	Flood forecasting	hydrological and meteorological dataset	-	DL(Federated Learning)	No	No	No	Pakistan
[57]	Water level monitoring and prediction	Images and video data	-	DL(CNN)	No	No	No	China
[65]	Two-step ML method for casualty prediction under emergencies	Emergency event and casualty data	Population density, infrastructure data, response time, event severity	ML/DL (Random Forest + Neural Network)	Yes	Yes	Yes	China

6.2. Artificial Intelligence in Prehospital Demand Forecasting, Response Time Prediction, and Resource Management

Forecasting and optimizing pre-hospital assets is crucial for maintaining high service standards and improving the overall patient experience. Proactive resource allocation helps address several systemic challenges, such as overcrowding in emergency departments, prolonged waiting times, and declining patient satisfaction due to delays in care. Minimizing the time between the initial call and medical intervention is essential to prevent adverse clinical outcomes. Traditionally, these complexities have been tackled using mathematical formalisms for analyzing discrete-event systems, which offer a strong framework for optimizing resources and strategic capacity planning. However, with the rise of AI, new opportunities are emerging to enhance these processes, providing more dynamic and predictive solutions for demand forecasting, response time prediction, and resource management.

6.2.1. Forecasting EMS Demand

Early detection of call volume provides valuable information for EMS personnel and dispatchers. The knowledge gained from analyzing the distribution of incoming call requests supports decision-making regarding the amount of resources required to meet future demand. In addition, fluctuations in EMS activity hinder the rational allocation of resources, with ambulance demand serving as a clear example [132]. Since ambulance demand fluctuates both spatially and temporally, depending on the day of week and particularly on the time of day, the most relevant information for ambulance deployment and workforce planning comes from insights obtained through accurate call volume forecasting [133] . Reliable demand forecasting enables the efficient allocation of ambulances and allows Emergency Medical Service (EMS) organizations to achieve their primary goal: ensuring adequate geographic coverage of service areas while minimizing response times to high-priority calls [63]. In addition, forecasting call volumes is critical, as it directly impacts ambulance dispatching and routing, helping to minimize response times and improve the quality and efficiency of emergency medical services. Similarly, demand dynamics are

strongly influenced by emergency events; therefore, accurately forecasting public health issues enables the reliable allocation of prehospital resources and enhances the overall performance of the emergency response system [134].

ML models have demonstrated remarkable capabilities to accurately forecast EMS demand by leveraging traffic flow, spatio-temporal data, and insights from social media [135]. In this context, [25] developed a stacking ensemble machine learning model to accurately predict the future occurrence of emergency calls as well as their geographic distribution, aiming to reduce the uncertainty associated with the variability in emergency service demand. The proposed ensemble model combines Decision Tree, Gradient Boosting Regression Tree, Light Gradient Boosting Machine, and Random Forest algorithms as base learners, while a Random Forest model serves as the meta-learner to produce the final estimation. The proposed approach outperforms single algorithms, achieving the best performance with an RMSE score of 0.9454. The model was trained on EMS data and incorporates contextual variables related to external factors such as weather conditions. Furthermore, an Explainable AI (XAI) method was applied to the model to highlight which external features most strongly influence the final predictions. Similarly, [63] proposed a decision support system to assist EMS managers in optimizing staffing and dynamic redeployment plans. The system is powered by a multi-layer perceptron (MLP) artificial neural network capable of predicting call volumes on a daily, hourly, and spatially distributed hourly basis. Features for training the prediction model were selected using machine learning techniques, particularly ensemble-based decision tree algorithms. In addition, unsupervised learning methods, particularly K-Means clustering, were employed to generate heterogeneous spatial clusters based on call locations and corresponding call volume densities, providing a clear visualization of hotspots where requests are most frequent.

Table 8. Artificial Intelligence for prehospital demand and response time forecasting and resource management

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[58]	Travel time	Ambulance Travel Data	ambulance GPS data route features temporal info	ML (RF)	No	No	No	Mexico
[59]	Number of daily ambulances	Ambulance Demand Data	historical ambulance demand weather temporal data	ML/DL (DNN, LightGBM)	No	Yes	No	Singapore
[60]	Number of ambulances needed	Ambulance Demand Data	historical ambulance dispatch temporal data	ML/DL (DNN, SVM)	No	No	No	Taiwan
[38]	Predicting EMS demand	Operational data	historical EMS calls temporal variables	ML (GMM, Naive Bayes, KNN, SVM, RF, Extra Trees, GB, AdaBoost)	No	No	No	Portugal
[61]	Forecasting EMS Time Series Data	Operational data	historical EMS calls hierarchical temporal structure	ML (SED, Poisson GLM, TSGLM, EDS, Ensemble)	No	No	No	UK
[62]	Prediction of demand volume	Operational data	historical EMS data temporal trends	DL (ANN)	No	No	No	Portugal
[63]	Forecasting the number of future calls to an emergency medical service (EMS) system, over time and across spatial zones	Operational data	historical EMS calls spatial temporal features	ML (K-Means, ANN)	No	Yes	No	USA

Table 9. Artificial Intelligence for prehospital demand and response time forecasting and resource management (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[64]	accuracy of overcrowding risk assessment in railway stations	EMS data	passenger flow data station characteristics temporal variables	DL/ML (ANN)	Yes	No	No	UK
[65]	prediction accuracy for casualties under emergency scenarios	Operational data	emergency scenario data hazard intensity contextual features	DL/ML (ANN, RF)	No	No	No	China
[66]	forecast accuracy for ICU healthcare resource demand	Operational data	epidemiological data hospital capacity temporal trends	DL/ML (GASVR, LSTM, BI-LSTM, Transformer, Hybrids)	No	No	No	China
[67]	spatiotemporal evolution metrics of medical waste generation	Operational data	public health emergency data population activity temporal information	ML (DE)	No	No	No	China
[68]	COVID-19 transmission prediction accuracy	Meteorological data	weather factors google search data temporal variables	ML (ARIMA, TSCC)	No	No	No	Australia
[69]	optimization of emergency medical resource allocation	Epidemiological data	epidemic severity area heterogeneity medical resource capacity	-	No	No	No	China
[70]	EMS dispatch recommendation performance metrics	Operational data	hospital bed availability ambulance availability system state	ML (Recommendation system)	No	No	No	Taiwan
[71]	effect of PM2.5 on seasonal influenza transmission	Environmental data	PM25 concentration climate zone classification spatial data	-	No	No	No	Australia

6.2.2. Operational Pre-Hospital Resource Management

In France and globally, EMS systems are facing mounting pressure due to a consistent rise in operational volume. This escalation directly correlates with an increased reliance on Emergency Medical Transport (EMT), thereby driving up overall demand. A critical observation regarding ambulance activity dynamics is its temporal fluctuation; demand is neither stable nor linear. This volatility leads to operational imbalances, where a surplus of EMT may sit idle during low-activity periods, while a deficit of units occurs during surges in emergency needs. Consequently, the adoption of optimization frameworks is vital to rationalize both human capital and financial expenditures. The initial stage of optimizing EMT deployment involves geographic mapping and the prediction of localized demand. To this end, [60] designed a flexible framework based on a predictive model of annual demand disaggregated into 2 km sectors, allowing for the spatial visualization of EMT requirements through a Geographic Information System (GIS).

In a similar vein, [59] developed a predictive model using a boosted random forest algorithm. Their approach integrated territorial attributes with short-term demand (the preceding 7 days) and long-term history (the previous 30 days) to forecast daily requirements. To ensure the interpretability and accountability of the model’s outputs, they implemented eXplainable AI (XAI) methodologies. Their findings revealed significant

correlations between demand levels and variables such as monthly and weekly trends, regional specificities, and the demographic proportion of individuals aged over 50.

Taking a more mathematical approach, [136] proposed a framework based on an inhomogeneous Poisson process. Their objective was to minimize the systemic risk associated with specific EMT configurations by evaluating call probabilities per locality, transit durations, and the total volume of available units. Such a methodology facilitates maximum coverage while mitigating global risk, ultimately promoting a more sustainable use of resources.

Similarly, research conducted in Mexico by [58] utilized Google Maps[®] to estimate transit times. However, they noted that standard navigation services are optimized for civilian vehicles and fail to account for the priority status of EMT, which typically navigate traffic more rapidly. To refine these estimates, they trained a corrective model based on trip characteristics (origin-destination), time of travel, and temporal markers (day and month). This model predicts a specific correction factor, ranging from a slight or moderate decrease to an increase to be applied to the baseline Google Maps duration, ensuring more realistic operational planning.

Furthermore, [129] highlighted the challenges of understaffing during high-demand intervals, establishing a direct correlation between insufficient personnel and increased waiting times for emergency access. To address this, they utilized the Erlang-C formula to evaluate service levels and project future staffing requirements. Their framework incorporates demographic variables alongside specific performance metrics, such as mandated response times. Consequently, stricter response time targets necessitate a proportional increase in personnel density to maintain compliance.

Table 10. Artificial Intelligence for prehospital demand and response time forecasting and resource management (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[72]	forecasting accuracy of response times to environmental events	Disaster data	environmental indicators climate variables historical response data	-	No	No	No	USA
[73]	probabilistic COVID-19 response prediction accuracy	EMS data	epidemiological data intervention measures temporal information	-	No	No	No	South Korea
[74]	uncertainty modeling performance for double standard model	-	fuzzy rules expert knowledge uncertain parameters	-	Yes	No	No	Mexico
[75]	early warning detection accuracy for seasonal influenza	Epidemiological data	syndromic surveillance data temporal patterns	DL/ML (SVM, RF, XGBoost, LSTM)	No	No	No	China
[76]	pandemic preparedness modelling outputs	Epidemiological data	climatic factors epidemiological data mobility data	-	No	No	No	Canada
[77]	prediction accuracy for norovirus incidence	-	intervention scenarios syndromic surveillance data search query data temporal variables	DL (RF)	No	No	No	UK

Table 11. Artificial Intelligence for prehospital demand and response time forecasting and resource management (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[78]	emergency logistics delivery optimization metrics	Geospatial	demand uncertainty resource availability transportation network disaster severity resource availability demand distribution temporal stages	-	No	No	No	China
[79]	emergency resource allocation optimization metrics	-	65 years of age or older Median income Living below the poverty level Highway miles daily calls to the EMS number and the daily dispatches of EMS vehicles time, location (place), and the nature of the medical complaint (incident)	DL/ML	No	No	No	China
[80]	forecast accuracy metrics for ambulance demand	EMS data	-	-	No	No	No	USA
[81]	early alert detection metrics for territorial COVID-19	EMS data	-	ML (PDE)	No	No	No	Italy
[82]	forecasting accuracy for emergency call volumes	EMS data	-	ML (AR)	No	No	No	Egypt
[83]	model selection performance for hospital asset demand	-	-	DL/ML (ANN, RNN, ANFIS)	No	No	No	Morocco
[84]	death or need for organ support	EMS ePCR data	Age, ACVPU-voice, ACVPU-pain, ACVPU-unresponsive, O2	NLP/DL/ML (SVM, ANN, GM, LR)	No	Yes	No	UK
[85]	mean absolute error and R ² for hourly EMS dispatches reduction in RMSE	EMS dispatch data	weather and temporal variables	DL/ML (RF, GLM, XGBoost, MARS, GAM)	No	Yes	No	USA
[86]	predicting events/hour	emergency data	weather and temporal variables	ML (XGBoost, AdaBoost)	No	No	No	Italy

6.3. Artificial Intelligence in Prehospital Emergency Triage and Risk Stratification

In the context of a mass casualty incident, where available resources are overwhelmed, a triage system becomes crucial for prioritizing life-threatening patients who require urgent prehospital care. The immediate needs of the population during such critical events place significant pressure on emergency decision-makers, who must quickly assess and prioritize the most critical cases to ensure that resources are allocated effectively. In this scenario, leveraging a triage system powered by AI is essential for rationalizing resource allocation, optimizing outcomes, and enhancing overall care for the affected community during disasters and health emergencies. Triage systems are designed to categorize, classify, and prioritize injured individuals based on their clinical needs, prognosis, urgency of treatment, and the resources available. In the context of EMS, triage of injured individuals or patients with illness is often initiated either over the telephone or by the rescue team upon their arrival at the scene of the incident.

6.3.1. Emergency Call Qualification and Prioritization Based on Audio Signal Analysis

Audio signals constitute the primary raw data within the EMS ecosystem. Utilizing these signals directly, rather than relying on clinical narratives, offers significant advantages in mitigating the risk of information loss or the introduction of biases inherent in transcription algorithms. Through mathematical formalisms rooted in digital signal processing, it is possible to isolate salient acoustic features (e.g., fundamental frequency (F_0), formants, Mel-Frequency Cepstral Coefficients (MFCC), jitter, shimmer, prosody, and rhythmic patterns) that support decision-making. In pre-hospital settings, such features are crucial for the early detection of cardiovascular emergencies, including strokes and out-of-hospital cardiac arrests (OHCA), as well as for assessing the patient’s emotional state and identifying critical physiological biomarkers.

In this context, [130], working with EMS 29 and 35 in Rennes, France, leveraged signal processing techniques to extract a comprehensive set of acoustic variables, such as fundamental and formant frequencies, vocal flicker, noise-to-harmonic ratios, and vocal interruptions. Their research involved a comparative analysis of binary logistic regression, random forests, and neural network architectures to detect OHCA. The findings indicated that the random forest model was the most effective, achieving an AUROC of 74.9% (with a 95% confidence interval). This suggests that acoustic analysis serves as a vital diagnostic complement to traditional verbatim analysis during emergency triage. Following a similar trajectory, [131] engineered an AI framework designed to categorize the levels of anxiety and cooperation exhibited by callers during cardiac arrest incidents. Their system evaluates the audio stream and assigns a classification across a five-point scale, ranging from "very stable and cooperative" (Level 1) to "uncontrollable and hysterical" (Level 5). Such a model is highly relevant for EMS applications, as it provides dispatchers (e.g., Medical Regulation Assistant (MRA), Emergency Medical Dispatcher (EMD), General Medical Dispatcher (GMD)) with real-time insights, allowing them to adapt their communication strategies to better stabilize and guide the caller through life-saving protocols.

Table 12. Artificial Intelligence in Prehospital Emergency Triage and Risk Stratification

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[87]	Department ID with highest propability	Triage Records Data	telehealth patient data	DL (BI-LSTM)	No	No	Yes	China
[88]	Preventable ED admission	Hospital admission Records	character embeddings	DL (RF, DT, XGBoost)	No	No	Yes	UK
[89]	Anaphylaxis event	Medical safety data	hospital patient data	NLP/ML (LSTM, Transformer, DT)	No	No	Yes	USA
[90]	Ambulance dispatch prioritisation	Dispatch Data	clinical history	NLP/ML (BERT, GPT, DT)	No	No	Yes	Australia
[91]	Undertriage	Call Trancripts	medical product reports	ML (RF, XGBoost)	No	No	Yes	Japan
[92]	Subsequent event	EMS Data	traffic crash reports	NLP/DL (FastText)	No	Yes	Yes	Finland
[93]	Prehospital Activated Major Trauma	Dispatch Data	features	NLP/ML (TF-IDF, Naive Bayes)	No	No	Yes	Taiwan
[94]	Emotional content	Voice Signals	telephone triage data	TV (SVM)	No	No	No	Taiwan
[95]	Critical care	EMS Critical Care Data	patient info call metadata	ML (LR)	No	No	No	South Korea
			EMS non-conveyance decisions					
			patient data					
			emergency call text features					
			patient vitals					
			call audio and speech features					
			patient vitals					
			prehospital data					

Table 13. Artificial Intelligence in Prehospital Emergency Triage and Risk Stratification(Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[96]	Admission	Dispatch-to-Hospital Data	prehospital patient vitals clinical indicators	ML (XGBoost)	No	No	No	Sweden
[97]	Probability of association	ED Data	EMS and ED records linking features	ML (LR)	No	No	No	USA
[98]	ICD-10	EMS records	prehospital emergency text records NLP features	NLP/DL (DNN, Transformer, BI-LSTM)	No	No	No	China
[100]	Patient admissions to ED	Triage Records Data	triage notes patient clinical info	NLP/DL (Transformer)	No	Yes	No	France
[101]	Emergency Reason Categories	Voice Signals	call audio textual features	NLP/DL (BERT)	No	No	No	Spain
[102]	Emergency calls severity	Voice Signals	call features multimodal data	NLP/DL (BERT)	No	No	No	France
[103]	Orientation upon leaving the emergency room	Hospital Admission Data	paramedic assessment patient demographics vital signs	ML (RF,GB, XGBoost)	No	No	No	Canada
[104]	Life-Threatening Event Classification	Dispatch Data	call transcripts textual features dataset shift indicators	NLP/DL (DNN, BERT)	No	No	No	Spain
[105]	Life-Threatening Level, Emergency System Jurisdiction	EMS Data	call transcripts multimodal features temporal patterns	NLP/DL (DNN, BiLSTM, BERT)	No	No	No	Spain
[107]	ICU admission	ED Data	patient demographics vitals initial assessment	NLP/ML (XGBoost)	No	Yes	No	Australia
[108]	Emergency calls' reason	EMS Data	emergency call transcripts textual data	NLP/DL (GPT-2)	No	No	No	France

6.3.2. Emergency Call Qualification and Prioritization Based on Operator Clinical Narratives

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Textual records generated during emergency calls represent an invaluable data repository. These clinical narratives document critical information such as the call’s context, geographic location, patient demographics, primary complaints, and the dispatcher’s preliminary assessment. By synthesizing these notes with assigned priority levels (e.g., P0–P4), researchers can develop automated triaging systems capable of distinguishing between life-threatening emergencies (P0) and non-urgent cases (P4). For example, [98] framed this challenge as a multi-class classification problem. By utilizing a convolutional neural network (CNN) trained on dispatcher-entered text, they were able to successfully categorize patient records into four distinct clusters based on pre-hospital data.

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These clusters encompass a diverse range of pathologies. Group 1, which represents the highest volume of pre-hospital documentation, covers ubiquitous conditions such as out-of-hospital cardiac arrest, cephalalgia, ankle injuries, and acute alcohol intoxication. Conversely, Group 4 constitutes a smaller subset of observations, focusing on conditions like hypoglycemia, seizures, and drug-related toxicities. The "reason for call" serves as a fundamental clinical biomarker for EMS dispatchers, often recorded well before the intervention of a regulating physician. [123] leveraged this variable at EMS 33 in France, utilizing a Large Language Model (LLM) to categorize call motives during the COVID-19 pandemic. Their study analyzed over 790,000 clinical notes from the Gironde region, reserving nearly 40,000 records for model validation and hyper-parameter optimization.

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The resulting model achieved accuracy scores ranging from 0.78 for generic categories (based on a massive training set of 412,218 entries) to 0.99 for specific focal neurological deficits (with 5,951 entries). These metrics were validated using a 95% confidence interval and a bootstrap-estimated bias correction. However, in environments where clinical scores are absent, supervised learning becomes unfeasible, necessitating a rigorous labeling phase to map observations to specific clinical classes. Furthermore, the presence of high-frequency "stop words", terms that lack significant semantic weight can impede algorithmic efficiency [124], requiring specialized preprocessing.

To address these limitations, Named Entity Recognition (NER) is employed to isolate pivotal medical concepts within the corpora, such as symptoms, patient age, therapeutic history, and contact details. Beyond clinical extraction, NER is indispensable for the de-identification of clinical narratives [125], ensuring compliance with stringent patient data protection regulations. A notable application by [126] utilized anonymized 911 dispatch reports to train NER models focused on severe trauma. While their approach demonstrated high sensitivity (Recall = 0.87), it struggled with specificity due to a substantial false-positive rate (Precision = 0.06), highlighting the ongoing challenge of precision in automated emergency text analysis.

6.3.3. On-site triage during mass casualty incidents

While remote prioritization via telephone prepares rescue teams to respond effectively upon arrival, on-site triage plays a crucial role in the stratification of patients requiring Intensive Care Units (ICU) services. This process allows the rescue team to efficiently prioritize patients based on their clinical needs observed at the scene, while accounting for available medical resources. To address this challenge, the implementation of intelligent triage systems powered by ML algorithms is essential to support the rescue team’s decision-making process.

For instance, a recent study integrates AI algorithms into networked unmanned aerial vehicles (UAVs), utilizing the 5th generation mobile network to ensure seamless signal transmission between the UAV components and the rescue team’s data stream. The AI-powered triage system perceives the incident environment in real time, employing detection algorithms to recognize and analyze body posture in order to assess the initial condition of injured individuals [127].

In these critical situations, training rescue teams with cutting-edge training protocols is indispensable. To address this challenge, AI is employed to generate disaster scenario simulations. This system immerses the rescue team in real-world conditions, enhancing their on-site response and emergency management skills [128].

Table 14. Artificial Intelligence in Prehospital Emergency Triage and Risk Stratification (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[109]	Hourly No. of admissions	ED Data	patient demographics vital signs historical admissions	ML (XGBoost, RF, LightGBM)	No	No	No	France
[110]	Emergency Severity Index	Triage Records Data	GPT-4 outputs patient data emergency scenario	LLM (GPT-4)	No	No	No	Israel
[111]	Injury type categories	Free-text ED records	free-text ED data clinical notes NLP features	LLM (GPT-4)	No	No	No	Italy

Table 15. Artificial Intelligence in Prehospital Emergency Triage and Risk Stratification(Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[109]	Hourly No. of admissions	ED Data	patient demographics vital signs historical admissions GPT-4 outputs	ML (XGBoost, RF, LightGBM)	No	No	No	France
[110]	Emergency Severity Index	Triage Records Data	patient data emergency scenario free-text ED data	LLM (GPT-4)	No	No	No	Israel
[111]	Injury type categories	Free-text ED records	clinical notes NLP features	LLM (GPT-4)	No	No	No	Italy
[112]	Documentation analysis	Chatbot Data	generative AI outputs prehospital care data clinical text	LLM (GPT-3.5)	No	No	No	USA
[113]	Emergency Detection	Free-text medical communication scenarios	patient description medical context call text	LLM (Llama)	No	No	No	USA
[114]	Triage and Dispatch Prioritization	Ambulance Demand Data	patient symptoms location information	LLM (GPT-4)	No	No	No	USA
[115]	Diagnostic errors	MIMIC-III-ED	patient age symptoms clinical text	LLM (ClinicalBERT)	Yes	No	No	USA
[116]	Accuracy of GPT-generated diagnoses	EHRs Data	clinical case description patient symptoms	LLM (GPT-3.5, GPT-4)	No	No	No	USA
[117]	Evaluate LLM performance in dispatch and triage	EMS Data	triage scenarios clinical guidelines LLM outputs	LLM (GPT-4)	No	No	No	USA
[118]	Emergency severity index Prediction	EMS Data	patient symptoms emergency severity rules	LLM (GPT-4)	No	No	No	Turkey
[119]	COVID-19 Diagnosis through Speech and Voice Analysis	Voice Signals	speech voice features acoustic signals	ML (SVM)	No	No	No	Italy
[120]	Event classification accuracy on social media posts	Seismic data	Text images captions temporal metadata	NLP (ViT)	No	No	No	China
[121]	Global event detection, sentiment and anomaly detection accuracy	-	Textual data temporal metadata geographic information	DL/ML (CNN)	No	No	No	Australia
[122]	Earthquake emergency knowledge extraction accuracy	-	Text, seismic, location	NLP/DL/ML (BI-GAT)	No	No	No	China

6.4. Artificial Intelligence for patient-centered prehospital critical care prognostication

The primary objective of the EMS call center is to deliver clinically appropriate interventions tailored to the patient's condition within an optimal timeframe, particularly in the context of disasters and health emergencies. These outcomes result from patient-centric, real-time decision-making performed by specialized emergency dispatchers. To support these decisions, synthesizing various data streams is crucial, including information triaged by operators (e.g., MRA, EMS, GP) and detailed clinical data from on-site teams (e.g., vital signs, clinical observations, and verbatim reports). Integrating such comprehensive data ensures the highest standards of patient safety and care quality. By leveraging this dataset, AI-powered predictive frameworks can be developed to anticipate the most suitable clin-

ical interventions in disaster or health crisis situations. These systems act as a proactive safeguard, assisting operators in refining their responses, mitigating risks, and preventing the loss of critical opportunities for the patient during high-pressure events.

6.4.1. Prediction of Patient Pathology

Zhang et al. [139] designed a multilayer neural network architecture to differentiate between confirmed stroke cases (encompassing both ischemic and hemorrhagic types) and stroke mimics, such as transient ischemic attacks. In fact, identifying strokes during a telephone interview remains a significant clinical challenge. Initial assessments by first responders often struggle to isolate objective neurological indicators from subjective perceptions. While Magnetic Resonance Imaging (MRI) is the definitive gold standard for diagnosis, its utility is hampered by limited availability and a stark supply-demand imbalance. Given that an MRI scan typically requires 30 minutes and the presence of a specialized radiologist, an AI-driven system capable of filtering genuine thrombolysis alerts from mimics holds substantial strategic value for emergency resource management.

Building on this logic, [161] developed a predictive model focusing on large vessel occlusion to facilitate ultra-rapid, patient-specific management. Such a tool could play a vital role in directing patients to the most appropriate facilities and allowing highly specialized surgical teams to prepare for their arrival under reduced stress. Ultimately, this predictive capability could prevent redundant inter-hospital transfers and ensure that patients are immediately funneled toward competent neurovascular centers.

From a similar theoretical standpoint, Traumatic Brain Injuries (TBIs) present a spectrum of care needs ranging from routine observation to immediate neurosurgical intervention, such as for subdural hematomas. The scarcity of specialized facilities capable of managing severe TBIs necessitates an exceptionally precise referral process. Accurate pre-hospital triage is essential to mitigate the overcrowding of specialized trauma centers, prevent the under-utilization of local resources, optimize the deployment of transport vectors, and minimize the geographical displacement of patients’ families.

Aligned with these objectives, [143] proposed a predictive diagnostic model for intracranial hemorrhage. Their algorithm, trained on data from the EMT, synthesizes a multi-dimensional dataset including demographics, consciousness levels (via the Glasgow Coma Scale), vital signs, pupillary reactions, injury mechanisms (high vs. low energy), secondary epilepsy, associated trauma, and clinical progression. Utilizing an eXtreme Gradient Boosting (XGB) framework, the model achieved an AUROC of 78%. These findings illustrate the significant potential of boosted tree algorithms in enhancing pre-hospital neurological diagnostics.

Table 16. Artificial Intelligence for patient-centered prehospital critical care prognostication

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[137]	New Injury Severity Score	Trauma Data	trauma registry data patient features	ML/DL (LR, RF, DNN)	No	No	Yes	Sweden
[138]	Transfer of patients to an inappropriate hospital	Patient Records	prehospital patient data cardiovascular indicators	ML (SVM, XGBoost, DT)	No	No	No	South Korea
[139]	Stroke mimic	Call Transcripts	prehospital patient data triage features resuscitation activity	DL (MLP, DNN)	No	No	Yes	China
[141]	Cerebral prognosis score	Recuscitation Data	features interaction metrics	DL (DNN)	No	No	No	Japan

Table 17. Artificial Intelligence for patient-centered prehospital critical care prognostication

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[142]	Acute coronary syndrome	Clinical Data	patient symptoms vitals ECG data	ML/DL (SVM, MLP, RF, LR)	No	No	Yes	Japan
[143]	Traumatic intracranial hemorrhage	Triage Records Data	prehospital patient data imaging features	ML (XGBoost, RF, SVM)	No	No	Yes	Japan
[144]	Out-of-hospital cardiac arrest	Voice Signals	call phonetic features audio signals OHCA	ML/DL (LR, RF, DNN)	No	No	Yes	France
[145]	Coronaragraphie	Care Workflow Data	workflow data prehospital interventions	ML/DL (FCN, LR, XGBoost, SVM)	No	No	No	USA
[146]	Stroke	Clinical Data	patient symptoms vital signs temporal variables	ML (RF, SVM, XGBoost)	No	No	Yes	Japan
[147]	Large Vessel Occlusion	Stroke Data	prehospital patient data vitals demographics	ML (LR, SVM, RF)	No	No	No	Japan
[148]	Out-of-hospital cardiac arrest	Dispatch Data	drone availability OHCA location constraints	ML/DL (LR, DNN)	No	No	No	Canada
[149]	Died at 1 month	Cardiac Arrest Outcome Data	patient initial rhythm prehospital data	ML/DL (LR, SVM, RF, DNN)	No	No	No	Japan
[150]	Stroke Delays - Admission	Stroke/Transport Data	patient clinical data transport time	ML (SVM)	No	No	No	China

6.4.2. Anticipating In-Transit Clinical Deterioration and Intensive Care Requirements

During the critical window of on-site stabilization or patient transfer to a medical facility, the risk of rapid clinical decline is a significant concern. The deployment of specialized medical teams, such as the SMUR (Service Mobile d’Urgence et de Réanimation), is vital for mitigating these risks and providing immediate life-saving interventions. To enhance this process, [159] created a predictive framework designed to foresee out-of-hospital cardiac arrest in severely injured victims of road traffic accidents during their evacuation. By utilizing Random Forest architectures trained on clinico-biological data from a massive cohort of 460,521 patients, the study achieved high predictive performance, with an AUROC of 95% and sensitivity/specificity values reaching 92%. This level of accuracy provides a robust tool for identifying high-risk patients before they reach the hospital.

Similarly, [95] developed a neural network-based approach using pre-hospital metrics to estimate the probability that a patient will require a critical care unit upon arrival at the emergency department. This methodology is instrumental in detecting early indicators of clinical instability, thereby facilitating the proactive dispatch of mobile intensive care units. Given the scarcity of high-level medical resources like SMUR teams, such predictive modeling ensures a more strategic allocation of assets, optimizing the medical response for those in the greatest need.

6.4.3. Prognostic Modeling Based on Pre-Hospital Clinical Outcomes

The utilization of pre-hospital intervention data including patient demographics, the frequency of defibrillations, pharmacological treatments, and the specific timing of telephonic or on-site rescuer contact, has enabled the development of advanced prognostic tools. For instance, in cases of suspected Road Traffic Accidents (RTA), research conducted

by [141] led to the creation of a predictive framework for neurological outcomes, ranging from minimal sequelae to persistent vegetative states or mortality.

By employing a three-layer neural network, the researchers achieved a high level of predictive accuracy, with an AUROC performance reaching 96%. This model functions effectively as a "digital twin," allowing for the simulation of various intervention scenarios by adjusting key variables, such as the delay before the first rescuer contact or the timing of initial defibrillation and medication. Consequently, this computational approach confirms critical operational objectives, specifically demonstrating the vital importance of minimizing response times and accelerating the delivery of initial shocks to improve neurological survival rates.

Table 18. Artificial Intelligence for patient-centered prehospital critical care prognostication (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[151]	1-month survival	Survival Data	OHCA incident data patient demographics	ML (RF)	No	Yes	No	Sweden
[152]	Acute aortic syndrome	Triage Records Data	patient symptoms vital signs imaging data	ML (LR)	No	No	No	France
[153]	Out of Hospital cardiac arrest	Voice Signals	OHCA incident data prehospital interventions	ML/DL (DT, RF, XGBoost, DNN)	No	No	No	USA
[155]	Electro-mechanical dissociation classification	ECG data	pulse signals patient vitals temporal patterns	ML (SVM)	No	No	No	USA
[156]	Cardiac arrest	EHRs Data	patient vitals clinical parameters	ML/DL (RF, LR, LSTM)	No	No	No	Taiwan
[157]	Mortality	Trauma Data	prehospital trauma patient records medical	ML (LR, RF, XGBoost)	No	No	No	USA
[158]	Stroke	Helpine Call Data	helpline call transcripts patient info	NLP/DL (CNN, RNN)	No	No	No	Denmark
[159]	Cardiac arrest during transport	Transport	patient vital signs transport data trauma info	ML/DL (SVM, DNN)	No	No	No	Japan
[160]	Heart attack	Nationwide Cardiac Data	nationwide prehospital database patient demographics	ML (XGBoost, GB)	No	No	No	South Korea
[161]	Stroke	Stroke Data	clinical data NLP/DL (LSTM, Transformer)	-	No	No	No	China
[162]	Sepsis, heart attack or cardiac arrest	Audio Data	caller descriptions incident context textual features	ML/DI (DT, DNN, XGBoost)	No	No	No	South Africa
[163]	Survival Prediction	Triage Records Data	remote patient data vital signs incident description	ML/DL (ANN, RF)	No	No	No	South Korea
[154]	accuracy in recognizing cardiac arrest from emergency calls	Audio data	emergency call audio text features	ML (ANN)	No	No	No	Denmark
[164]	Estimation of the number of victims in a mass-casualty incident (MCI)	Operational data	smart glasses video visual features	DL (ANN)	No	No	No	Thailand

Table 19. Artificial Intelligence for patient-centered prehospital critical care prognostication (Cont.)

Ref	Outcome	Data	Context variables	Applied ML Techniques	UQ	XAI	CCA	Country
[165]	triage accuracy of Google Bard and ChatGPT	Clinical data	patient scenario descriptions triage rules clinical context cardiac arrest incident locations	ML/DL/NLP (GPT)	No	No	No	Spain
[140]	-	Clinical data	temporal patterns demographic data	NLP (ANN)	No	No	No	South Korea

7. Discussion

7.1. Most Used Algorithms for Use Cases in the EMS

This section outlines the prevailing AI methodologies employed in the development of decision support systems within the EMS landscape. The data visualized in our analysis, derived from a rigorous selection of literature, offers a comprehensive look at algorithmic preferences over time. As demonstrated in Figure. 9, there is a clear chronological shift in how researchers model structured data, text, audio signals, and sequences in prehospital settings.

A primary observation from this longitudinal data is the sustained preference for classical ML over DL architectures. A significant majority of the reviewed studies prioritize traditional ML, with a specific focus on decision-tree-based methodologies. While neural networks saw a surge in research activity between 2018 and 2022, the Random Forest algorithm established itself as the dominant choice by 2022. Its widespread adoption is largely credited to its consistent performance across a variety of EMS-specific applications.



Figure 9. Most used algorithms over time.

The popularity of ensemble methods, such as Random Forest and Extreme Gradient Boosting (XGBoost), can be attributed to several factors. These algorithms are notably resilient to overfitting, a critical advantage in regression and classification tasks within the medical domain. Furthermore, they tend to yield high-quality results with minimal initial tuning and exhibit strong performance when handling the noisy data characteristic of emergency environments. Crucially, decision-tree models are generally more interpretable than neural networks, satisfying the transparency requirements essential for clinical decision support.

In parallel, Support Vector Machines (SVM) and Artificial Neural Networks (ANN) also experienced a resurgence in interest during 2022. This trend is likely driven by their proven effectiveness in managing high-complexity operations, including multimodal forecasting, natural language processing, and the analysis of visual data from medical imaging [50,166].

7.2. Challenges of Data Quality, Completeness, and Algorithmic Bias in EMS AI

The operational environment of EMS is defined by high-pressure scenarios, the necessity for synchronized interventions, and critical time constraints. Within this framework, ensuring data quality and completeness remains a formidable obstacle. For example, acoustic analyses of prehospital patient recordings have demonstrated substantial interference between speech and environmental noise, alongside fragmented voice signals. Furthermore, discrepancies and gaps are prevalent in dispatch records, including triage logs, prehospital intervention reports, and emergency documentation. These observations align with existing literature; notably, a recent assessment of EMS data integrity found that the average completeness for triage records, care reports, and emergency forms reached only 52.3%, 70.3%, and 57.3%, respectively [167]. Similarly, an evaluation of the Michigan MI-EMSIS database revealed alarmingly high rates of missingness for vital variables, with body temperature absent in over 90% of documented cases [168]. Such low levels of data saturation significantly compromise the dependability of clinical decision-making and the validity of subsequent research. This data heterogeneity, stemming from either EMS software architectures or variations in recording practices among personnel, introduces systematic biases into the analysis of emergency incidents.

As the implementation of ML and AI accelerates in EMS—specifically for dispatch enhancement, pre-triage, and acuity prediction—the issue of algorithmic bias has become a paramount concern [169]. This phenomenon occurs when a system consistently generates inequitable, irregular, or faulty results, typically mirroring the structural inequalities present in the training datasets. These biases are rooted in the information fed to AI models during their development, directly influencing how the algorithms are parameterized. If the training data is skewed, the resulting model calibrations will amplify these disparities, potentially leading to discriminatory outcomes based on ethnicity, gender, socioeconomic background, or cultural factors. Research has established a significant correlation between algorithmic biases and the inherent cognitive biases of human actors [170]. In essence, the outputs of these systems are not merely a product of software design but are deeply shaped by the limitations of historical records, which often fail to encompass all causal relationships and are colored by human interpretation. These observations emphasize the vital role of human supervision in AI-driven decision-support systems to ensure that recommendations remain contextualized, equitable, and compliant with the ethical mandates of emergency medicine.

7.3. Uncertainty, Transparency and Explainability in AI for Emergency Medical Services

In the high-pressure environment like EMS, ensuring the transparency and explainability of AI systems is fundamental to maintaining clinical safety, trust, and equitable care. Transparency involves the degree to which various actors—including clinicians, dispatchers, and regulatory bodies—can grasp the internal mechanics of an AI system, such as its architecture, input data, and underlying logic. Explainability, however, extends this concept by ensuring that algorithmic outputs can be interpreted and justified through human-centric reasoning, which is particularly vital for systems designed to assist in time-critical, real-time interventions [171].

Historically, "natural" statistical approaches, such as the Classification and Regression Tree (CART) methodology, provided intrinsic explainability. These models utilize a hierarchical structure where variables are processed sequentially, allowing a human observer to easily follow the decision path from the root node to the final prediction [172]. Conversely, modern "black box" frameworks, like deep learning and neural networks, do not align with human-readable logic. Their intricate layers and weighted nodes interact in complex ways that bypass traditional clinical decision rules, potentially hindering institutional oversight, professional trust, and accountability. Despite the critical nature of EMS operations, recent

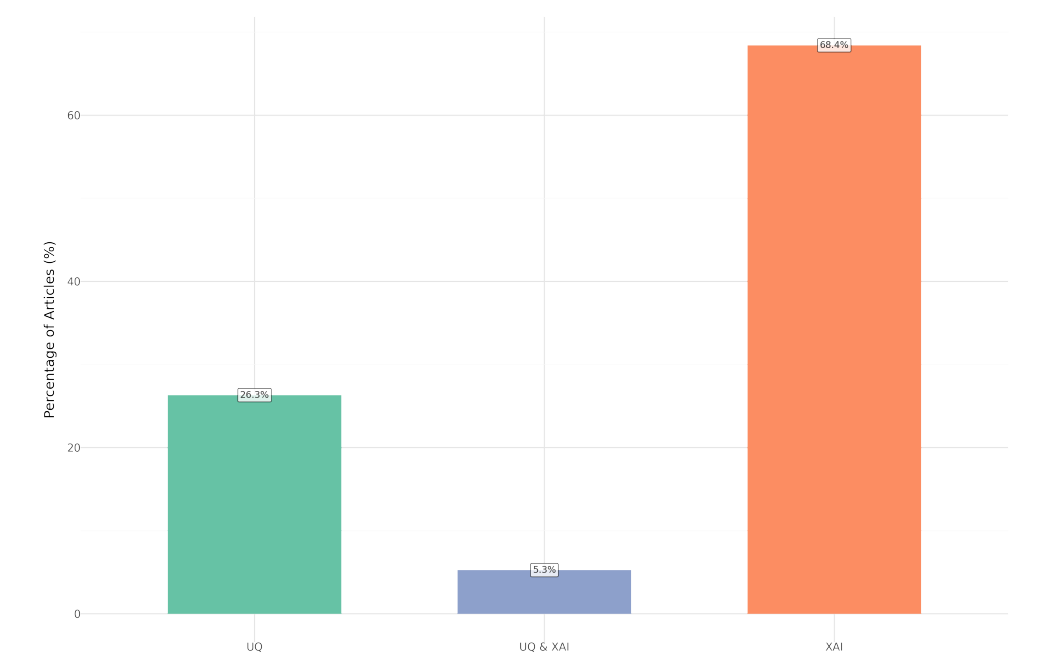


Figure 10. Distribution of papers addressing uncertainty, transparency, and explainability in AI systems for EMS.

literature indicates that explainable AI (XAI) remains infrequently utilized in predictive tools for triage, dispatching, or pre-hospital management. While algorithmic precision has seen significant improvements, a majority of current studies fail to incorporate post-hoc diagnostic techniques—such as SHAP, LIME, or counterfactual analysis—and rarely involve medical practitioners in verifying the model’s rationale. As illustrated in Figure 10, the opaque nature of machine learning continues to obstruct clinical integration, as responders cannot adequately defend algorithmic suggestions to their peers or patients [173].

Analysis of existing research shows that while interpretability frameworks are occasionally discussed, comprehensive transparency is often lacking. Many studies focus predominantly on predictive metrics, treating explainability as a superficial or secondary concern. Given the rapid operational pace and high-risk profile of tasks like dispatching and field triage, AI models must be as credible as they are accurate. A failure to clarify the

reasoning behind an algorithmic recommendation can result in clinician doubt, delayed responses, or the unintended reinforcement of bias. Consequently, in the EMS sector, explainability is no longer an optional feature; it is a core requirement for the ethical and secure implementation of AI technologies.

7.4. Data Privacy Issues and Poor Generalization

The concept of AI generalization remains ambiguous, presenting significant legal and ethical dilemmas. Currently, the restricted ability of AI models to generalize serves as a primary hurdle in the pursuit of Artificial General Intelligence (AGI). This bottleneck is rooted in a lack of resilient learning algorithms capable of functioning across varying domains, alongside a deficit in engineered systems that demonstrate the autonomous and adaptive qualities inherent in human cognition [174]. While prior research emphasizes that external validation using standardized reference datasets is essential for confirming model generalizability [175], accessing such data in the medical field is heavily restricted. Strict adherence to data-privacy mandates is required to safeguard individual subjects, particularly when sharing datasets with the broader data-science community for model verification [176].

In practice, Electronic Health Records (EHRs) are comprised of highly sensitive information characterized by personally identifiable markers. The presence of Protected Health Information (PHI), which is often interspersed throughout unstructured clinical notes, complicates the sharing of EHRs for academic purposes and hinders their utility in achieving data-driven clinical goals [177]. Regulatory frameworks—such as Health Insurance Portability and Accountability Act (HIPAA) in the United States, the General Data Protection Regulations (GDPR) in the European Union, and Commission nationale de l’informatique et des libertés (CNIL) standards in France—dictate that identifiable patient data be restricted to the primary clinical team. Access for external researchers is permitted only with explicit patient authorization or under specific legal exemptions.

To facilitate data utilization across all development phases, the extraction of PHI is mandatory [178]. This objective is typically met through anonymization techniques, which offer a robust pathway for regulatory compliance [179]. However, pseudonymization is frequently preferred to maintain the clinical context; this involves identifying PHI elements and substituting them with generic descriptors (e.g., "[SURNAME]", "[LOCATION]"). This de-identification process is a technical prerequisite for the development and deployment of generalized clinical models. Furthermore, the historic difficulty in obtaining annotated data to train these de-identification tools has created a technical stalemate. Recently, the adoption of Large Language Model (LLMs) has helped resolve this issue. LLMs possess the extensive linguistic knowledge required to effectively de-identify EHRs, thereby facilitating the training of robust AI systems for clinical use [180]. This evolution may eventually streamline the validation of AI models on shared datasets, leading to more versatile and reliable medical AI.

7.5. Knowledge Acquisition in Prehospital Decision-Making

Regarding current practices in the prehospital setting, knowledge acquisition methods have become more sophisticated as a result of the increasing complexity of questions faced by prehospital decision-makers. Traditional decision-making approaches, though effective in simpler scenarios, are often inadequate in addressing the dynamic challenges presented in prehospital care. In this context, the integration of AI-driven tasks such as recommendation[181], forecasting[182], and detection [183] has become essential in providing the tools needed to support decision-making and mitigate the uncertainty inherent in emergency situations. Figure 11 illustrates the distribution of these AI-driven knowl-

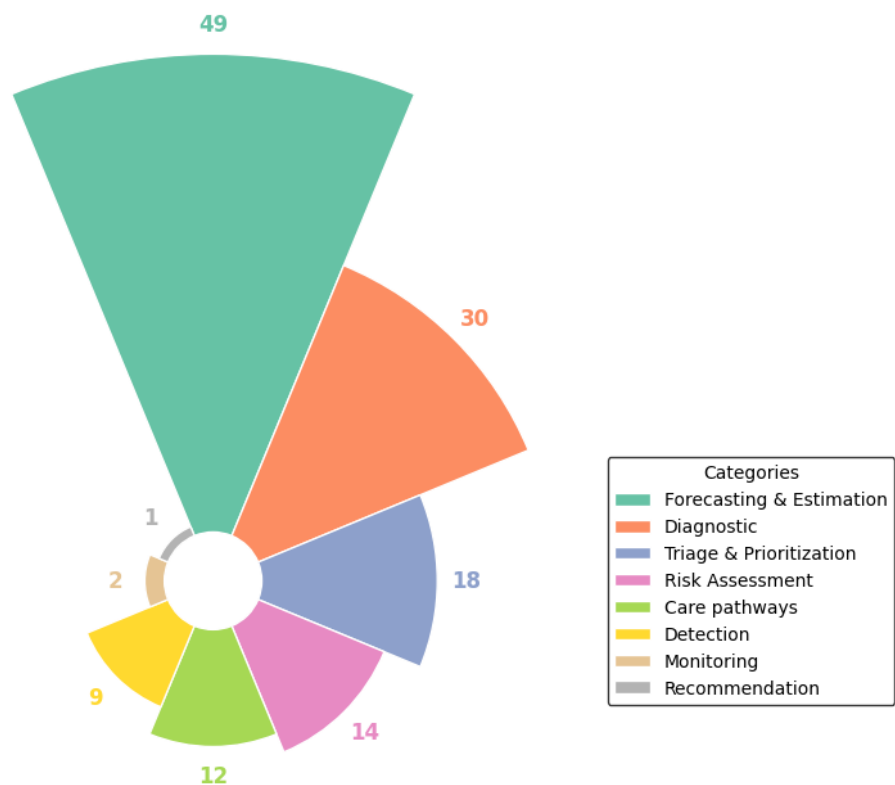


Figure 11. Distribution of AI tasks for knowledge acquisition in the prehospital sector.

edge acquisition methods. Notably, the Forecasting Estimation domain stands out as the most prevalent, with a significant rise in the number of articles focused on this area. This increase can be attributed to the types of questions commonly posed by prehospital decision-makers, such as: ‘What will happen next?’ or ‘What actions are needed to achieve a desired outcome?’. These types of predictive questions are critical for managing uncertainty and planning effectively in the prehospital sector. In second place, we find Diagnosis, which plays a key role in clinical decision-making. This domain highlights the utility of AI in assisting healthcare professionals with understanding patients’ clinical histories and predicting their future medical trajectory. By leveraging AI, decision-makers can make more informed judgments regarding the patient’s immediate needs. For other, less-represented AI task domains, the corresponding questions may be less frequently posed, or concerns about legal and ethical implications may limit their broader adoption in prehospital decision-making.

7.6. *Agentic AI Systems and Model Context Protocol (MCP) in EMS Workflow*

The integration of LLMs into EMS workflows represents a significant advancement in supporting optimal prehospital care and reducing the risk of adverse outcomes for critically ill patients [184]. However, standalone LLMs often lack sufficient contextual grounding to generate responses precisely tailored to a patient’s clinical condition, potentially leading to suboptimal or incomplete recommendations. Furthermore, their effectiveness is constrained by fragmented EMS infrastructures and strict data governance requirements, which may limit access to relevant information distributed across hospital and prehospital databases, thereby restricting context-aware decision-making.

In high-stakes environments where heterogeneous information including operational, environmental, and patient-level data must be processed in real time by dispatchers, standardized approaches for LLMs integration and structured ingestion of heterogeneous data

and tools are essential. Recent advances in the MCP enable secure and interoperable connections to multiple data sources and auxiliary systems [185]. MCP facilitates the orchestration of smaller, specialized LLMs by a coordinating supervisory model within a multi-agent architecture. Such a design supports functional decomposition and asynchronous task execution under an explicit orchestration layer, thereby enhancing scalability, modularity, and computational efficiency in complex EMS workflows.

For a given task, a “collaboration crew” of specialized sub-agents can be instantiated to generate coordinated and coherent outputs, reflecting a structured form of distributed reasoning analogous to multidisciplinary human teamwork. The integration of agentic AI systems powered by MCP-based tool orchestration holds significant promise for improving the quality, reliability, and consistency of EMS decision support, ultimately contributing to improved patient outcomes [186]. Moreover, MCP enables agents to access comprehensive and dynamically updated knowledge bases, including triage algorithms, dispatcher protocols, and other relevant operational or clinical resources. This capability supports the generation of recommendations that approach expert-level reasoning while maintaining real-time assistance for dispatch personnel.

7.7. Adversarial Attacks on AI Systems and Human Guarantees

Recent advances in AI systems represent significant progress toward more general and multi-task AI agents. However, despite their high performance, these systems remain vulnerable to adversarial attacks [50]. AI models, including LLMs, operate within complex infrastructures composed of multiple interconnected components. Even when a model is properly trained, vulnerabilities may arise from corrupted libraries, insecure data pipelines, or malicious query routing mechanisms. Adversarial attacks can manipulate both inputs and outputs, potentially compromising system reliability and safety. Such risks are particularly critical in high-stakes domains like EMS, where erroneous or manipulated outputs may negatively affect patient outcomes.

To mitigate these threats, robust defense strategies must be implemented at both technical and organizational levels. In particular, human-in-the-loop AI systems provide an additional safeguard by ensuring transparency, auditability, and explicit handling of uncertainty [187]. Integrating structured human oversight strengthens system reliability and supports compliance with clinical governance and patient safety requirements. Combining technical robustness with human guarantees is essential to enhance trust, safety, and resilience against adversarial threats.

7.8. Opportunities of AI in Emerging EMS Systems

The existence of a universal EMS system remains a complex subject due to the global-scale lack of a standardized and unified model, despite a strong convergence toward shared clinical principles. However, socio-economic inequalities constrain many regions in adopting established EMS protocols due to country-level realities, leading to a significant disparity in development between systems. In Low- and Middle-Income Countries, this problem is mainly exacerbated by a low national Gross Domestic Product (GDP), which creates significant infrastructural and systemic barriers.

In this context, AI integration represents a transformative opportunity for emerging EMS systems to override traditional structural limitations. For instance, AI applications can assist these systems in emergency triage, patient prognosis, and real-time diagnosis, while ensuring continuous patient outcome monitoring [188]. Furthermore, AI technology can support emotional detection and identify critical care needs through biomarker detection in the caller’s voice. Additionally, it enables the enhancement and correction of emergency voice signals, which are vital for identifying critical cases in high-stress or

noisy environments. Beyond clinical care, AI optimizes operational planning and resource allocation—a necessity both in disaster contexts and during non-strain periods to ensure maximum efficiency of limited assets.

Similarly, due to the lack of trained emergency dispatchers in many emerging regions, the use of AI can automate the emergency dispatch process. This automation could ensure a standardized, data-driven response to every call, relieving physicians and specialists of administrative burdens. By doing so, it makes them available for on-site pre-hospital care and specialized in-hospital interventions, where their expertise is most critical.

8. Conclusion

AI represents a major milestone toward resilient EMS systems under emergency conditions. The integration of ML-based approaches within EMS is becoming essential as health emergencies and disasters grow increasingly frequent and unpredictable. In this article, we provide the community with a holistic review of how AI can be integrated into EMS workflows to address health emergencies and disasters effectively.

EMS can be viewed as a complex socio-technical system positioned at the interface between the population, healthcare institutions, and public authorities. This strategic positioning makes EMS a key stakeholder for the deployment of advanced AI technologies in crisis response. AI integration in prehospital systems serves multiple functions: it supports routine operations and high-demand scenarios and may facilitate earlier identification of critical events with the potential to escalate into large-scale crises.

To assess the feasibility and current state of such integration, we analyzed studies investigating AI techniques in prehospital emergency services as well as AI applications in disaster and health emergency management. Publications screened were issued between 2015 and early 2025. The PRISMA methodology was applied to ensure rigorous selection of relevant studies. Articles were included based on their contribution to AI domains such as machine learning, deep learning, generative AI, and agentic AI, as well as their practical application within EMS in both routine and crisis contexts.

Our findings highlight the significant benefits of AI in disaster prediction, health emergency forecasting, environmental risk assessment, demand and resource allocation under emergency conditions, and prehospital triage, outcome prediction, and decision support in life-threatening situations. These applications demonstrate the transformative potential of AI across the entire emergency response continuum.

However, we observed a limited proportion of studies addressing uncertainty quantification, interpretability, and explainability within proposed algorithms. While several works apply post hoc explainability tools, only a few integrate intrinsic interpretability or uncertainty-aware mechanisms directly into model design. In cases where simpler models (e.g., decision tree-based or regression approaches) are considered interpretable due to reduced complexity, uncertainty estimation remains insufficiently addressed.

Future research should focus on developing data-efficient algorithms capable of providing domain-expert-oriented explanations while explicitly quantifying predictive uncertainty. Such advancements are essential to manage the increasing volatility and unpredictability of healthcare systems and related domains. Moreover, stronger efforts are required to enhance robustness against adversarial attacks and to systematically incorporate human-in-the-loop mechanisms, ensuring safety, accountability, and trust in AI-driven EMS systems.

Author Contributions: Conceptualization, K.L.E; methodology, K.L.E, O.E., Y.K; software, K.L.E, O.E., Y.K; validation, T.D. and J.C.; formal analysis, K.L.; investigation, K.L.E., O.E., and O.B.; resources, K.L.E., O.E.; data curation, O.E., K.L.E.; data ingestion K.L.E., Y.K; writing—original draft preparation, K.L.E; writing—review and editing, O.E., Y.K, J.C., O.B. and T.D.; visualization, K.L.E., O.E. and J.C.;

Funding acquisition, O.B., J.C. and T.D.; supervision, O.B.; project administration, T.D. and J.C. All authors have read and agreed to the published version of the manuscript.

Funding: This study is funded by the Interreg France–Switzerland 2024–2027 cooperation program. While funded by the project, the views expressed are those of the authors.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Most of the data supporting the findings of this study are included within the article. Regarding the data used to highlight the EMS distribution across the world, it will be made available by the authors upon request.

Acknowledgments: This work was supported by the GESICA project (Gestion des Situations de Crises sAnitaires), funded by the Interreg France–Switzerland 2024–2027 cooperation program. This project is led by Marie Louis Pasteur University (UMLP) and Geneva University Hospitals (HUG).

Conflicts of Interest: Author Thibaut Desmettre is affiliated with the Emergency Department, HUG Genève, Switzerland. The authors of the manuscript, Omar Elfahim, Kokou Laris Edjinedja, Yendouname Kantame, Johan Cossus, Oussama Barakat, and Thibaut Desmettre, declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence	
AUROC	Area Under the Receiver Operating Characteristic Curve	
AVP	Accident de la Voie Publique	
CRRC 15	Call Reception and Regulation Center	
CSNP	Unscheduled Care Centers	
DE	Emergency Dispatcher	
DSS	Decision Support System	
ED	Emergency Department	
EM	Emergency Medicine	
EMD	Emergency Medical Dispatcher	
EMS	Emergency Medical Service	
EMSCC	Emergency Medical System Call Center	
EMT	Emergency Medical Technicians	
GDPR	General Data Protection Regulation	
GP	General Practitioner	
GRU	Gated Recurrent Unit	1116
ICU	Intensive Care Unit	
LLMs	Large Language Models	
LSTM	Long Short-Term Memory	
MAS	Multi-Agent Systems	
MCP	Model Context Protocol	
ML	Machine Learning	
MRA	Medical Regulation Assistant	
NLP	Natural Language Processing	
OSNP	Other Specialized Non-Physician	
GDP	Gross Domestic Product	
PHI	Protected Health Information	
EHRs	Electronic Health Records	
AGI	Artificial General Intelligence	
CNIL	Commission nationale de l’informatique et des libertés	

NER	Named Entity Recognition	
RL	Reinforcement Learning	
RNN	Recurrent Neural Network	
SSE	Situation Sanitaire Exceptionnelle	1117
EMT	Emergency Medical Transport	
XAI	Explainable Artificial Intelligence	
CCA	Computational Complexity Analysis	

Appendix A

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Appendix A.1 Uncertainty Quantification methods in AI and Evaluation Metrics

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The decision-making process in Emergency Medical Services (EMS) is subject to high levels of uncertainty, often because operators must make decisions based solely on patients' complaints communicated over the phone. This challenge is further compounded when the EMS operator lacks access to the patient's historical medical data yet must still make rapid decisions to minimize response-time risks. In scenarios where an AI or machine learning (ML) system is designed to model call diagnosis and prioritization, quantifying uncertainty becomes a critical stage. Doing so enhances the robustness, interpretability, and reliability of ML-driven decisions. Various scientific tools can assist in developing techniques to address these challenges, typically classified into six categories: probabilistic methods, ensemble learning methods, sampling-based approaches, generative models, deterministic methods, and emerging hybrid techniques. While these proposed methods are robust for quantifying uncertainty in AI systems, without appropriate evaluation metrics, it is difficult to compare, select, or validate their effectiveness. We summarize in Table A1 the uncertainty quantification methods, their evaluation metrics, and some key characteristics.

Table A1. Summary of Major Uncertainty Quantification (UQ) Techniques in Machine Learning.

Category	Principle	Data Level	Knowledge Level	Typical Applications	Evaluation Metrics
Probabilistic Methods	Represent uncertainty using probability distributions (e.g., Bayesian Neural Networks, Variational Inference, Monte Carlo Dropout).	Yes	Yes	Predictive modeling in healthcare, finance, and autonomous systems.	Entropy, Confidence Intervals, Expected Calibration Error (ECE), Maximum Calibration Error (MCE), Brier Score, Prediction Interval Width (PIW).
Ensemble Learning Methods	Combine predictions from multiple models to estimate mean and variance.	Yes	Yes	Medical image analysis, autonomous navigation, adversarial robustness.	Variance between models, Disagreement metrics, Calibration metrics (ECE, MCE), Brier Score.
Sampling-Based Methods	Approximate posterior distributions via stochastic sampling (e.g., MC, HMC, SMC).	No	Yes	Bayesian optimization, model calibration, uncertainty-aware reinforcement learning.	Posterior distribution analysis, Credible Intervals, Effective Sample Size, Log Score, Continuous Ranked Probability Score (CRPS).
Generative Models	Learn data distributions using latent variable models (e.g., VAEs, GANs, Normalizing Flows).	Yes	No	Medical imaging, physics-informed modeling, anomaly detection.	Likelihood / Log-likelihood, KL-Divergence, Predictive Entropy, AUROC for OOD detection.
Deterministic Methods	Estimate uncertainty deterministically without sampling (e.g., Evidential Deep Learning, Quantile Regression).	No	Yes	Real-time or resource-constrained systems.	Prediction Intervals, Quantiles, Entropy, Coverage Probability, Sharpness metrics.
Hybrid Methods	Combine multiple UQ strategies (e.g., Bayesian-Ensemble, Physics-Informed Neural Networks, information-theoretic UQ).	Yes	Yes	Robotics, climate science, material discovery, scientific modeling.	Combination of metrics above: ECE, MCE, Brier Score, PIW, Log Score, CRPS, AUROC.

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